MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE KBL

* INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital multiplexer module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.0 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KBL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/40-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

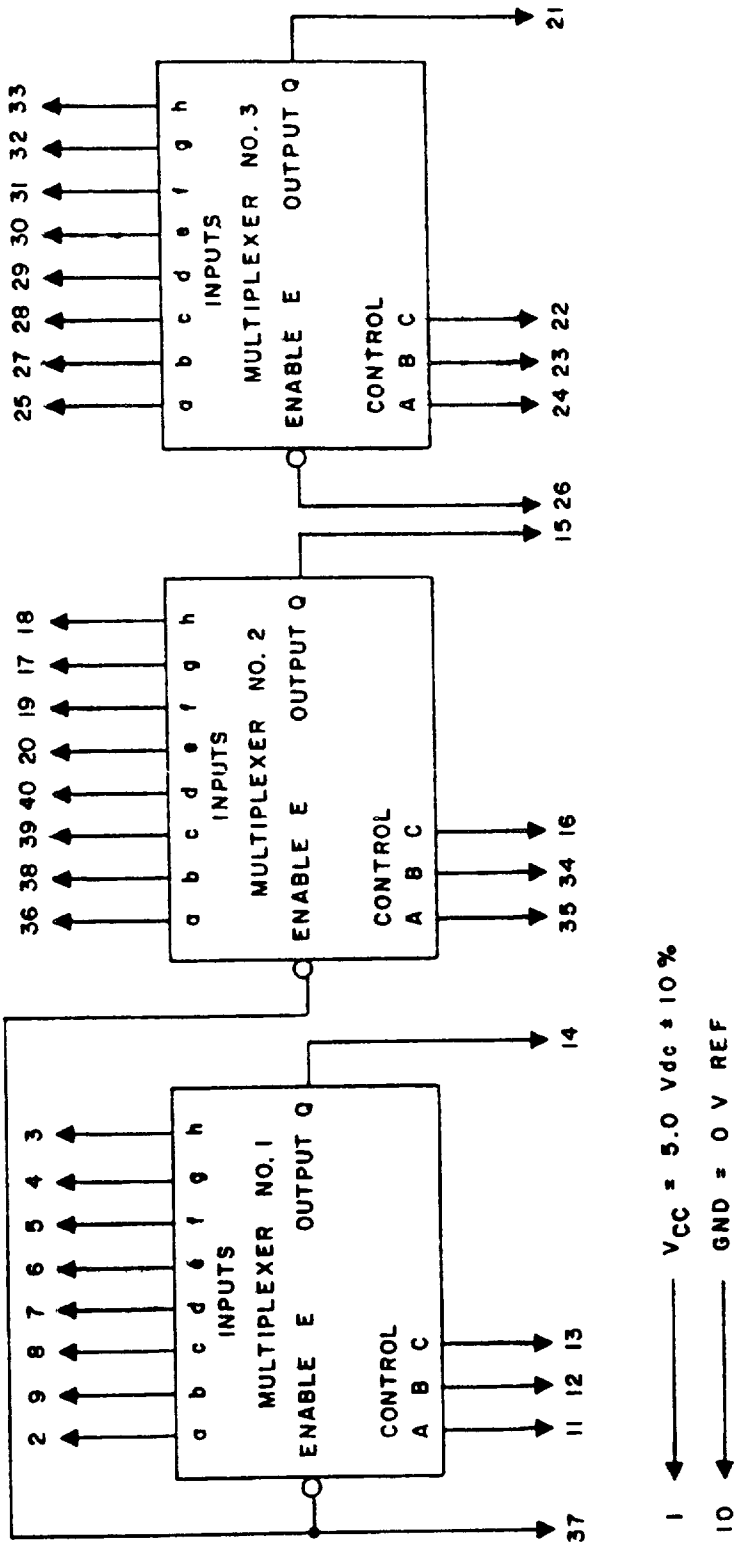


FIGURE 1. KBI module function diagram, multiplexer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	V_{OUT} (low-loaded)	III	S	0 01	0.6	0 01	0.6	0.01	0.6	0.01	0.6	V
B1	Input breakdown current (E1 & 2)	IV	S	0	2.0	0	2.0	0	2.0	0	2.0	mA
B2-B35	Input breakdown current (E3, A-C, and a-h)	IV	S	0	1 0	0	1.0	0	1.0	0	1.0	mA
C1-C35	Input clamp diode voltage	V	S	-0.4	-1 5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D48	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E72	Output voltage, low level	VI	X	0.00	0.30	0 00	0.30	0 00	0.30	0.00	0.30	V
F1	Input current, high level (E1 & 2)	VII	X	-10	40	-10	40	-10	40	-10	40	µA
F2-F35	Input current, high level (E3, A-C, and a-h)	VII	X	-10	20	-10	20	-10	20	-10	20	µA
G1	Input current, low level (E1 & 2)	VII	X	-0.01	-0 84	-0.01	-0 84	-0.01	-0.84	-0.01	-0.84	mA
G2-G35	Input current, low level (E3, A-C, and a-h)	VII	X	-0.01	-0.42	-0.01	-0.42	-0.01	-0 42	-0.01	-0.42	mA
H1-H3	Output short circuit current	VII	X	-5	-40	-5	-40	-5	-40	-5	-40	mA
J1	Power supply current, low level	VIII	S	10	40	10	40	10	40	10	40	mA
J2	Power supply current, high level	VIII	S	10	40	10	40	10	40	10	40	mA
K	Isolation	IX	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
L1	Input capacitance (E1 & 2)	X	Q	4	30	4	30	4	30	4	30	pF
L2-L35	Input capacitance (E3, A-C, and a-h)	X	Q	2	15	2	15	2	15	2	15	pF
M1-M3	Transition time (t_{TLH}), C_L = 50 pF	XI	S	5	60	5	60	5	70	5	70	ns
M4-M6	Transition time (t_{TLH}), C_L = 150 pF	XI	Q	10	110	10	110	10	120	10	120	ns

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N3	Transition time (t_{THL}), C_L = 50 pF	XI	S	2	30	2	30	2	35	2	35	ns
N4-N6	Transition time (t_{THL}), C_L = 150 pF	XI	Q	5	60	5	60	5	65	5	65	ns
P1-P24	Propagation delay time (t_{PLH}), a-h, C_L = 50 pF	XI	S	10	95	10	95	10	95	10	95	ns
P25,P33,P41	Propagation delay time (t_{PLH}), E, C_L = 50 pF	XI	S	10	105	10	105	10	105	10	105	ns
P26-P32, P34-P40, P42-P48	Propagation delay time (t_{PLH}), E, C_L = 50 pF	XI	Q	10	105	10	105	10	105	10	105	ns
P49,P50, P52-P58, P60-P70, P72-P74, P76-P82, P84-P94, P96-P98, P100-P106, P108-P118 P120	Propagation delay time (t_{PLH}), A-C, C_L = 50 pF	XI	Q	10	110	10	110	10	110	10	110	ns
P51,P59,P71, P75,P83,P95, P99,P107, P119	Propagation delay time (t_{PLH}), A-C, C_L = 50 pF	XI	S	10	110	10	110	10	110	10	110	ns
P121-P144	Propagation delay time (t_{PLH}), a-h, C_L = 150 pF	XI	Q	10	110	10	110	10	110	10	110	ns
P145-P168	Propagation delay time (t_{PLH}), E, C_L = 150 pF	XI	Q	10	120	10	120	10	120	10	120	ns

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P169-P240	Propagation delay time (t_{PLH}), A-C, C_L = 150 pF	XI	Q	10	125	10	125	10	125	10	125	ns
Q1-Q24	Propagation delay time (t_{PHL}), a-h, C_L = 50 pF	XI	S	10	85	10	85	10	85	10	85	ns
Q25,Q33,Q41	Propagation delay time (t_{PHL}), E, C_L = 50 pF	XI	S	10	105	10	105	10	105	10	105	ns
Q26-Q32, Q34-Q40, Q42-Q48	Propagation delay time (t_{PHL}), E, C_L = 50 pF	XI	Q	10	105	10	105	10	105	10	105	ns
Q49-Q50, Q52-Q58, Q60-Q70, Q72-Q74, Q76-Q82, Q84-Q94, Q96-Q98, Q100-Q106, Q108-Q118, Q120	Propagation delay time (t_{PHL}), A-C, C_L = 50 pF	XI	Q	10	115	10	115	10	115	10	115	ns
Q51,Q59,Q71, Q75,Q83,Q95, Q99,Q107, Q119	Propagation delay time (t_{PHL}), A-C, C_L = 50 pF	XI	S	10	115	10	115	10	115	10	115	ns
Q121-Q144	Propagation delay time (t_{PHL}), a-h, C_L = 150 pF	XI	Q	10	100	10	100	10	100	10	100	ns
Q145-Q168	Propagation delay time (t_{PHL}), E, C_L = 150 pF	XI	Q	10	120	10	120	10	120	10	120	ns
Q169-Q240	Propagation delay time (t_{PHL}), A-C, C_L = 150 pF	XI	Q	10	130	10	130	10	130	10	130	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R	Power supply transient amplitude	XII	Q	0	400	0	400	0	400	0	400	mV
S1-S3	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	mV
T1-T3	Output crosstalk, low level	XIII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XIV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, H, or K are part of a sequence of tests, the order of testing shall be as follows: A, B, K, H where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through J and M through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4 5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25° ±5°C ambient.

4 5 3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through J and M through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through J and L through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6 1 Definitions. Definitions are as specified in MIL-STD-1331

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/40-1 supersedes part number M28787/40. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	219 mW (outputs not loaded)
Supply voltage	5.0 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

Input current, low level		
a-h, A-C, and E3		
(V_{IL} = 0.0 V)	-0.42 mA maximum	
(V_{IL} = 0.3 V)	-0.40 mA maximum	
E1 & 2		
(V_{IL} = 0.0 V)	-0.84 mA maximum	
(V_{IL} = 0.3 V)	-0.80 mA maximum	
Input current, high level		
a-h, A-C, and E3		
(V_{IH} = 2.4 V)	20 µA maximum	
(V_{IH} = 5.5 V)	1.0 mA maximum	
E1 & 2		
(V_{IH} = 2.4 V)	40 µA maximum	
(V_{IH} = 5.5 V)	2.0 mA maximum	
Output voltage		
Low level	0.00 V minimum, 0.30 V maximum	
High level	2.4 V minimum, V_{CC} maximum	
Output drive current maximum capability		
Low level (V_{OL} = 0.3 V)	4.0 mA	
High level (V_{OH} = 2.4 V)	-400 µA	
Propagation delay time	C_L = 50 pF	C_L = 150 pF
a-h to Q		
t_{PLH}	95 ns maximum	110 ns maximum
t_{PHL}	85 ns maximum	100 ns maximum
A-C to Q		
t_{PLH}	110 ns maximum	125 ns maximum
t_{PHL}	115 ns maximum	130 ns maximum
E to Q		
t_{PLH}	105 ns maximum	120 ns maximum
t_{PHL}	105 ns maximum	120 ns maximum

6.5 Module function. This module contains three low-power 8-input multiplexers. Each multiplexer receives eight input data lines (a through h), three input data control lines (A, B, and C) and an enable input line (E). The data line used is selected by the state of the data control inputs. This data is transferred to the output (Q) providing the enable input is active (low). When the enable is not activated, the output is low regardless of all other inputs. Multiplexers 1 and 2 have the enable input (E) common. Multiplexer 3 has a separate enable input (E).

6.5.1 Input to output relationship. Each 8-input multiplexer has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE II. Input to output relationship. 1/, 2/

Inputs												Output
Control			Enable	Data								
C	B	A	E	h	g	f	e	d	c	b	a	Q
X	X	X	H	X	X	X	X	X	X	X	X	L
L	L	L	L	L	X	X	X	X	X	X	X	L
L	L	L	L	H	X	X	X	X	X	X	X	H
L	L	H	L	X	L	X	X	X	X	X	X	L
L	L	H	L	X	H	X	X	X	X	X	X	H
L	H	L	L	X	X	L	X	X	X	X	X	L
L	H	L	L	X	X	H	X	X	X	X	X	H
L	H	H	L	X	X	X	L	X	X	X	X	L
L	H	H	L	X	X	X	H	X	X	X	X	H
H	L	L	L	X	X	X	X	L	X	X	X	L
H	L	L	L	X	X	X	X	H	X	X	X	H
H	L	H	L	X	X	X	X	X	L	X	X	I
H	L	H	L	X	X	X	X	X	H	X	X	H
H	H	L	L	X	X	X	X	X	X	L	X	L
H	H	L	L	X	X	X	X	X	X	H	X	H
H	H	H	L	X	X	X	X	X	X	X	L	L
H	H	H	L	X	X	X	X	X	X	X	H	H

1/ $Q = \overline{E} \cdot (ABCa+\overline{A}BCb+A\overline{B}Cc+\overline{A}\overline{B}Cd+AB\overline{C}e+\overline{A}\overline{B}\overline{C}f+A\overline{B}\overline{C}g+\overline{A}\overline{B}\overline{C}h)$
2/ X = Level does not affect output.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity.
Navy - EC

(Project 5963-N300)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

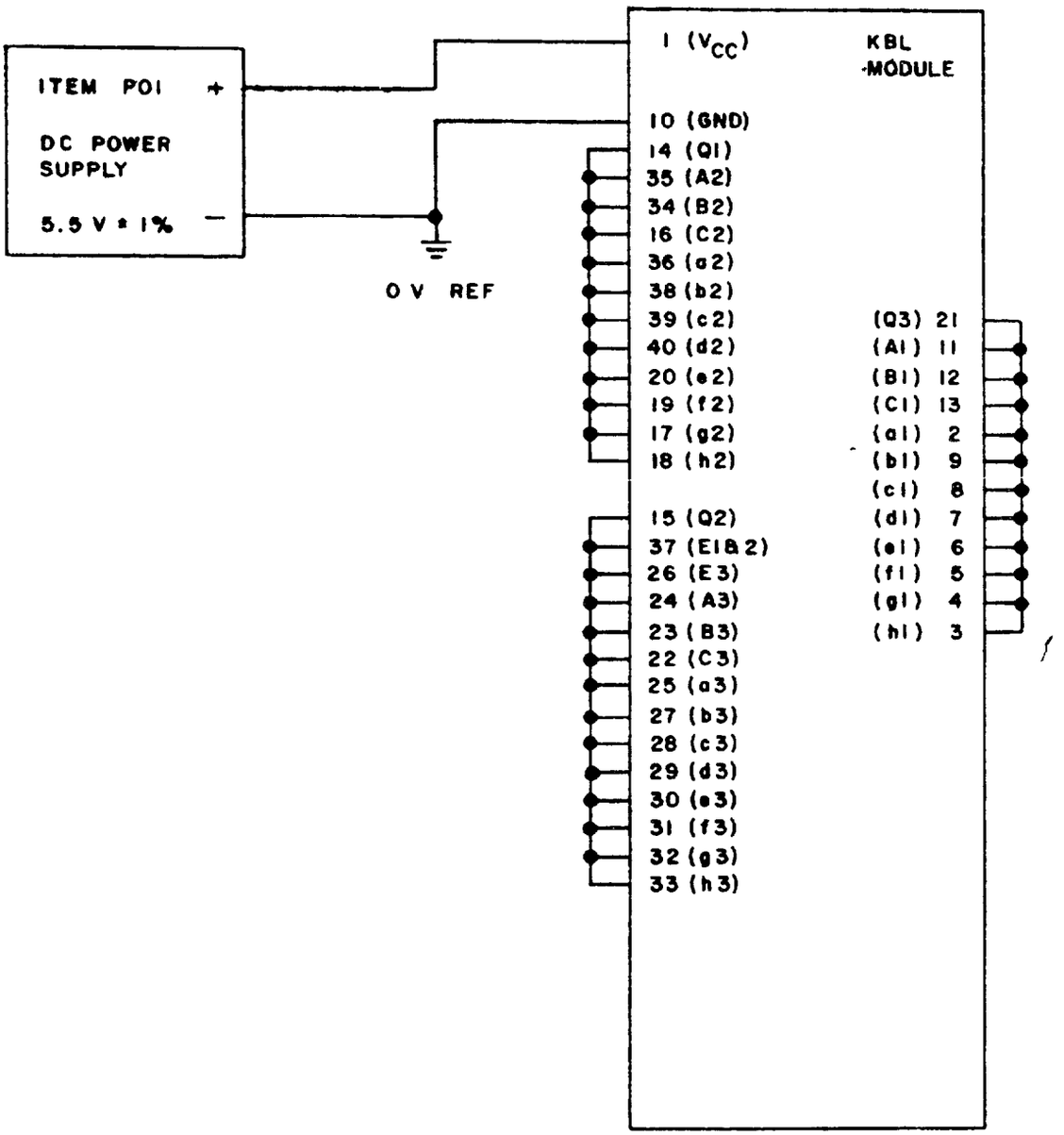
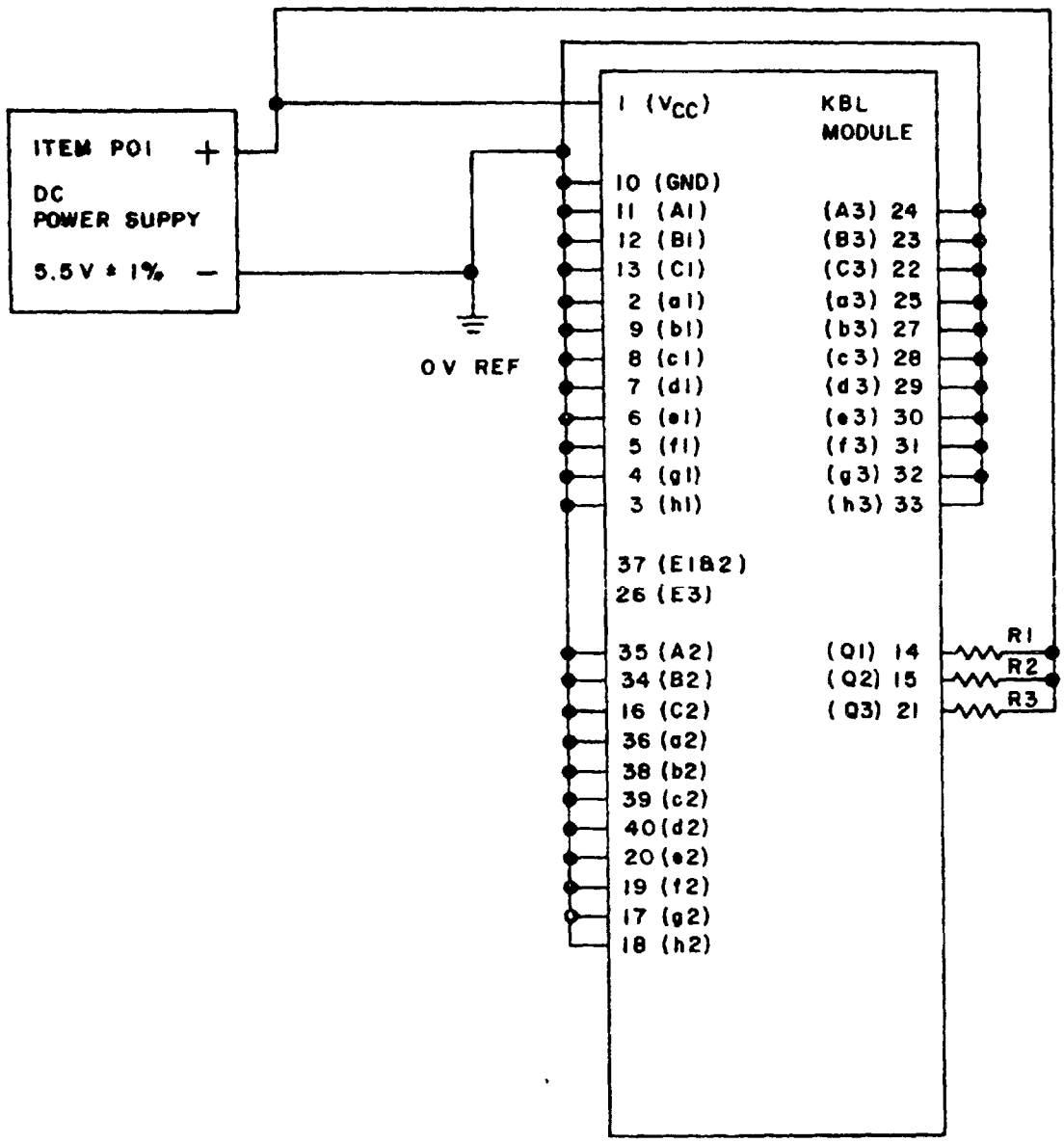


FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1-R3 = 1.33 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

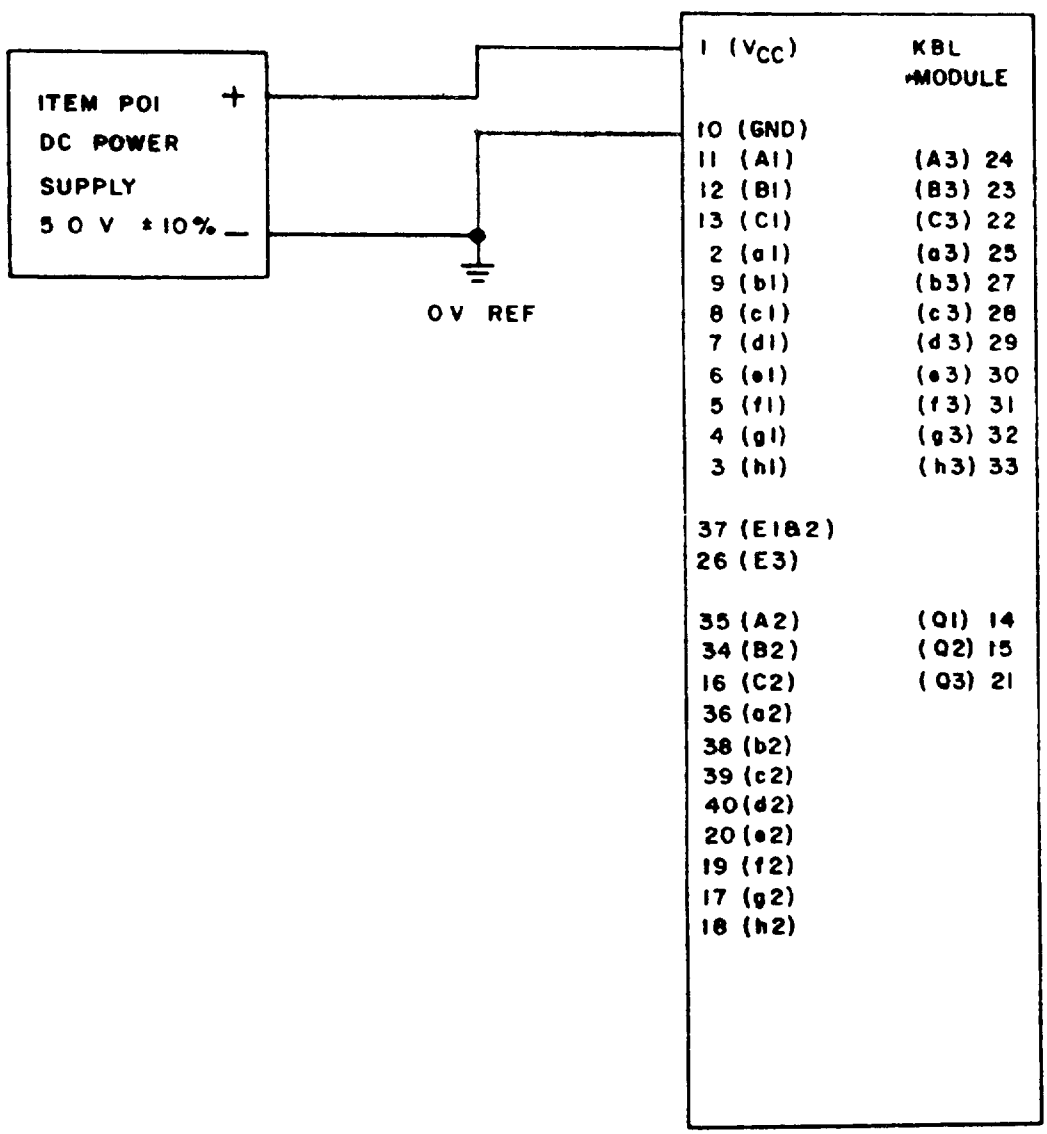


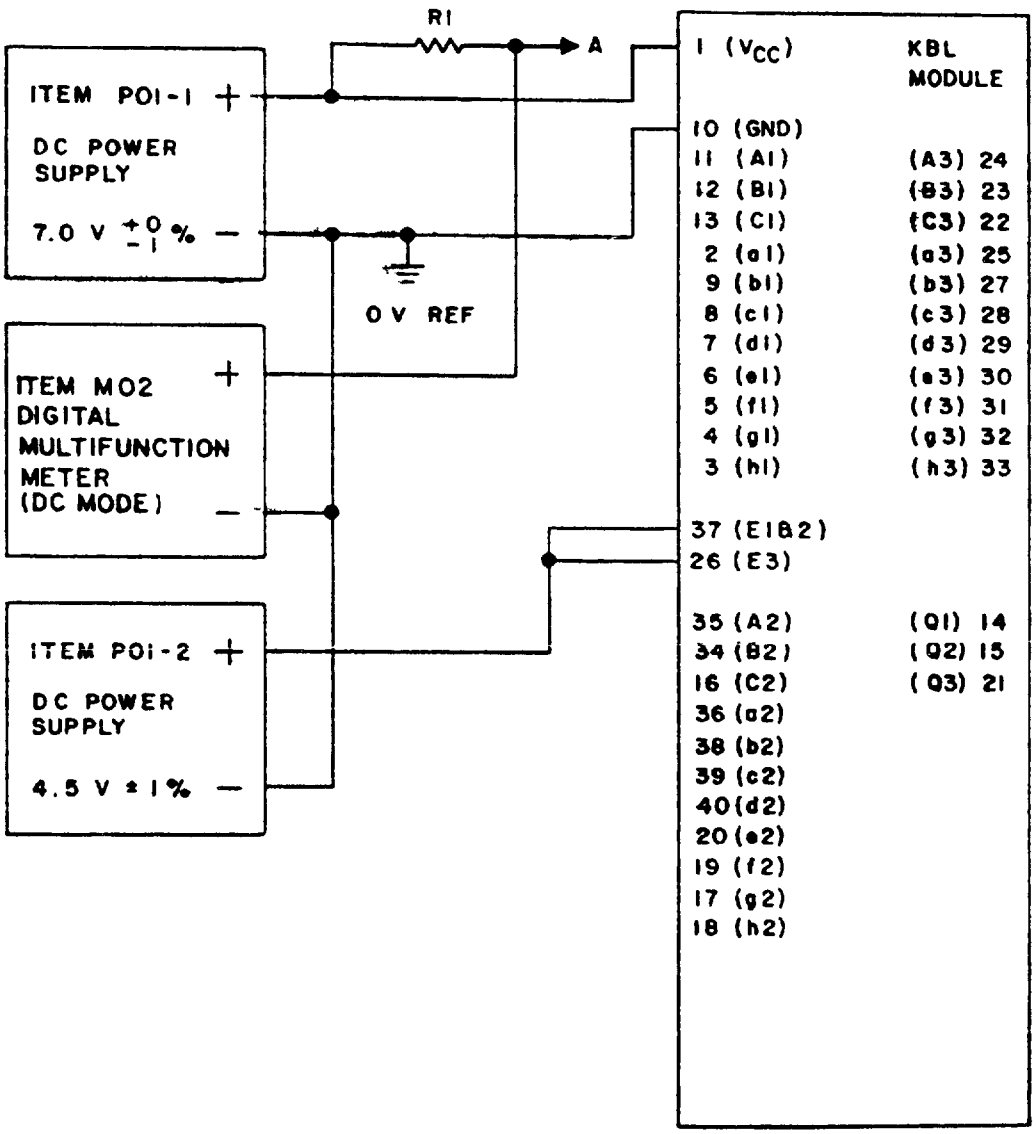
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE III. V_{OUT} (load-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	V_{OUT} (low-loaded)	14	0.01	0 6	V	5
A2	V_{OUT} (low-loaded)	15	0.01	0.6	V	5
A3	V_{OUT} (low-loaded)	21	0 01	0.6	V	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE R1 = 1.33 kΩ ±1%, 1/8 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test

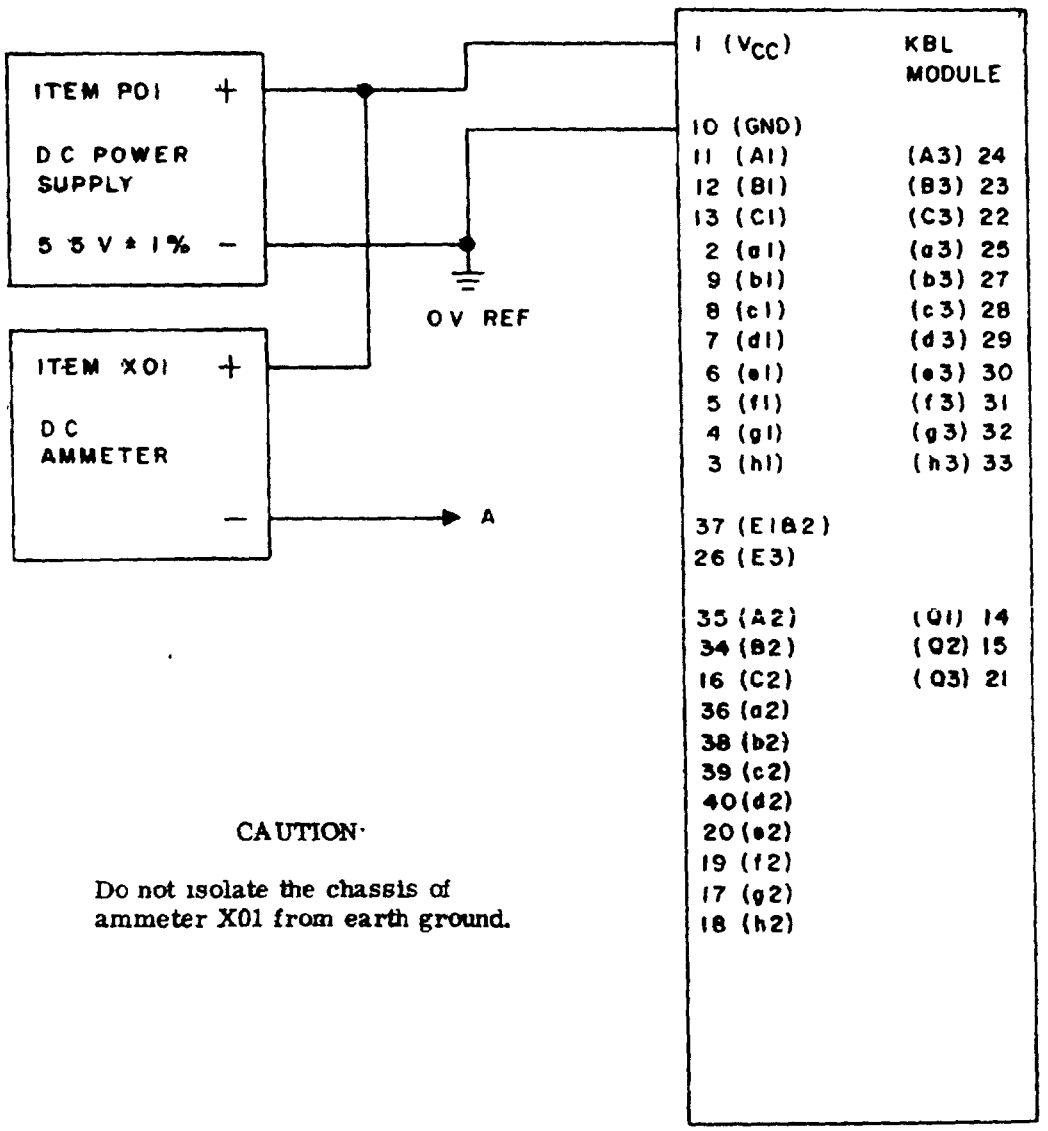
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
B1	E1 & 2 Input breakdown current	37	Open	11,12,13,35, 34,16	0	2.0	mA	6
B2	A1	11		37,12,13		1.0		
B3	B1	12		37,11,13				
B4	C1	13		37,11,12				
B5	a1	2	37	11,12,13				
B6	b1	9	37,11	12,13				
B7	c1	8	37,12	11,13				
B8	d1	7	37,11,12	13				
B9	e1	6	37,13	11,12				
B10	f1	5	37,11,13	12				
B11	g1	4	37,12,13	11				
B12	h1	3	37,11,12,13	Open				
B13	A2	35	Open	37,34,16				
B14	B2	34	Open	37,35,16				
B15	C2	16	Open	37,35,34				
B16	a2	36	37	35,34,16				
B17	b2	38	37,35	34,16				
B18	c2	39	37,34	35,16				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
B19	d2 Input breakdown current	40	37,35,34	16	0	1.0	mA	6
B20	e2	20	37,16	35,34				
B21	f2	19	37,35,16	34				
B22	g2	17	37,34,16	35				
B23	h2	18	37,35,34,16	Open				
B24	E3	26	Open	24,23,22				
B25	A3	24		26,23,22				
B26	B3	23		26,24,22				
B27	C3	22		26,24,23				
B28	a3	25	26	24,23,22				
B29	b3	27	26,24	23,22				
B30	c3	28	26,23	24,22				
B31	d3	29	26,24,23	22				
B32	e3	30	26,22	24,23				
B33	f3	31	26,24,22	23				
B34	g3	32	26,23,22	24				
B35	h3	33	26,24,23,22	Open				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE V Input clamp diode voltage test

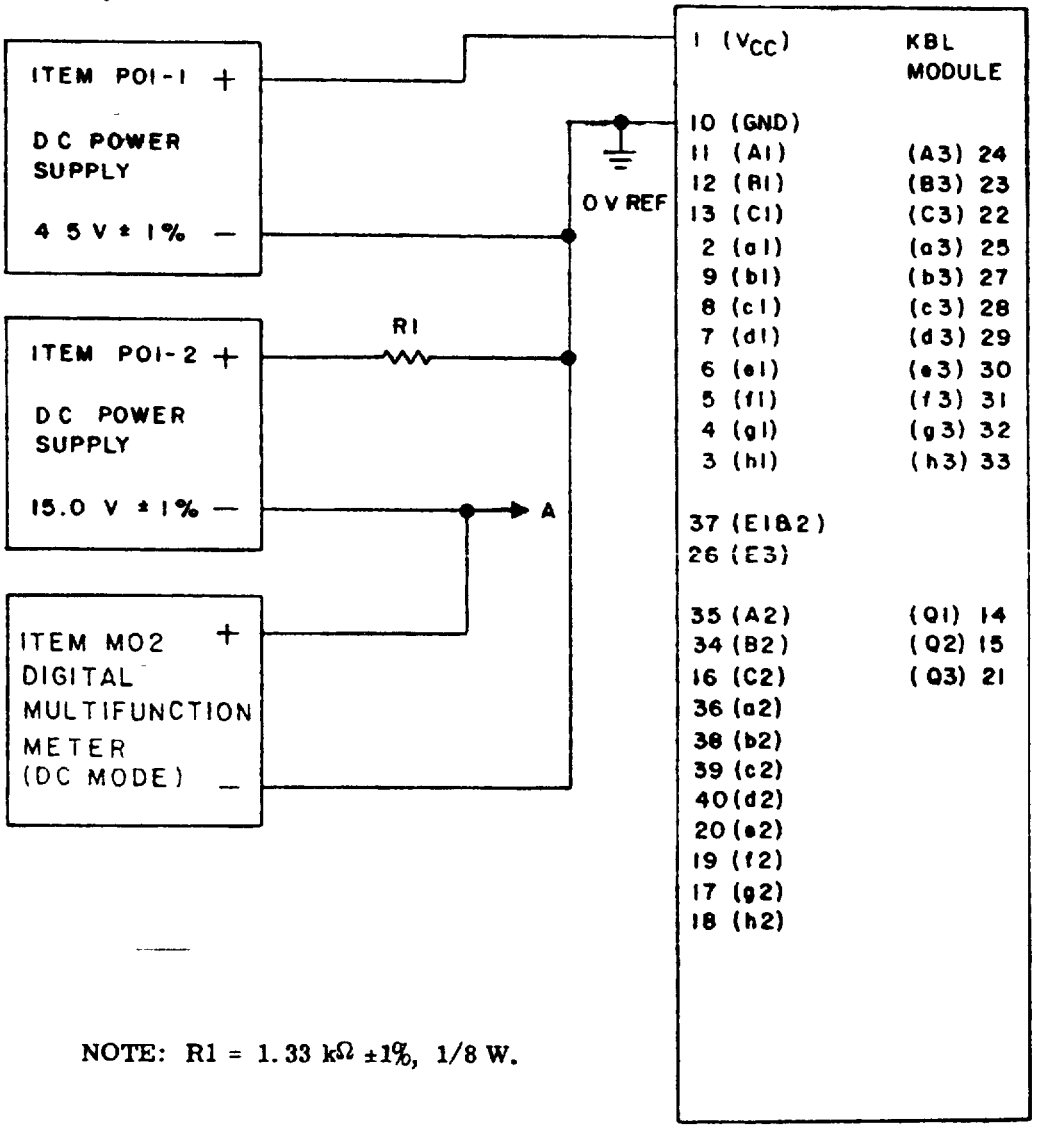
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	E1 & 2 Input clamp diode voltage	37	-0 4	-1.5	V	7
C2	A1	11				
C3	B1	12				
C4	C1	13				
C5	a1	2				
C6	b1	9				
C7	c1	8				
C8	d1	7				
C9	e1	6				
C10	f1	5				
C11	g1	4				
C12	h1	3				
C13	A2	35				
C14	B2	34				
C15	C2	16				
C16	a2	36				
C17	b2	38				
C18	c2	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C19	d2 Input clamp diode voltage	40	-0.4	-1.5	V	7
C20	e2	20				
C21	f2	19				
C22	g2	17				
C23	h2	18				
C24	E3	26				
C25	A3	24				
C26	B3	23				
C27	C3	22				
C28	a3	25				
C29	b3	27				
C30	c3	28				
C31	d3	29				
C32	e3	30				
C33	f3	31				
C34	g3	32				
C35	h3	33				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 1.33 kΩ ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test.

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D1	Q1 Output voltage, high level	11,12,13	37	14	2	9,8,7,6,5,4,3	2.4	4.5	V	8
D2	Q1	12,13	37,11		9	2,8,7,6,5,4,3				
D3	Q1	11,13	37,12		8	2,9,7,6,5,4,3				
D4	Q1	13	37,11,12		7	2,9,8,6,5,4,3				
D5	Q1	11,12	37,13		6	2,9,8,7,5,4,3				
D6	Q1	12	37,11,13		5	2,9,8,7,6,4,3				
D7	Q1	11	37,12,13		4	2,9,8,7,6,5,3				
D8	Q1	Open	37,11, 12,13		3	2,9,8,7,6,5,4				
D9	Q2	35,34,16	37	15	36	38,39,40,20, 19,17,18				
D10	Q2	34,16	37,35		38	36,39,40,20, 19,17,18				
D11	Q2	35,16	37,34		39	36,38,40,20, 19,17,18				
D12	Q2	16	37,35,34		40	36,38,39,20, 19,17,18				
D13	Q2	35,34	37,16		20	36,38,39,40, 19,17,18				
D14	Q2	34	37,35,16		19	36,38,39,40, 20,17,18				
D15	Q2	35	37,34,16		17	36,38,39,40, 20,19,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D16	Q2 Output voltage, high level	Open	37,35, 34,16	15	18	36,38,39,40, 20,19,17	2.4	4.5	V	8
D17	Q3	24,23,22	26	21	25	27,28,29,30, 31,32,33				
D18	Q3	23,22	26,24		27	25,28,29,30, 31,32,33				
D19	Q3	24,22	26,23		28	25,27,29,30, 31,32,33				
D20	Q3	22	26,24,23		29	25,27,28,30, 31,32,33				
D21	Q3	24,23	26,22		30	25,27,28,29, 31,32,33				
D22	Q3	23	26,24,22		31	25,27,28,29, 30,32,33				
D23	Q3	24	26,23,22		32	25,27,28,29, 30,31,33				
D24	Q3	Open	26,24,23, 22		33	25,27,28,29, 30,31,32				
D25	Q1	2	Open	14	11,12,13	37				
D26	Q1	9			12,13	37,11				
D27	Q1	8			11,13	37,12				
D28	Q1	7			13	37,11,12				
D29	Q1	6			11,12	37,13				
D30	Q1	5			12	37,11,13				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D31	Q1 Output voltage, high level	4	Open	14	11	37,12,13	2 4	4.5	V	8
D32	Q1	3		14	Open	37,11,12,13				
D33	Q2	36		15	35,34,16	37				
D34	Q2	38			34,16	37,35				
D35	Q2	39			35,16	37,34				
D36	Q2	40			16	37,35,34				
D37	Q2	20			35,34	37,16				
D38	Q2	19			34	37,35,16				
D39	Q2	17			35	37,34,16				
D40	Q2	18			Open	37,35,34,16				
D41	Q3	25		21	24,23,22	26				
D42	Q3	27			23,22	26,24				
D43	Q3	28			24,22	26,23				
D44	Q3	29			22	26,24,23				
D45	Q3	30			24,23	26,22				
D46	Q3	31			23	26,24,22				
D47	Q3	32			24	26,23,22				
D48	Q3	33			Open	26,24,23,22				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E1	Q1 Output voltage, low level	11,12,13	37	14	9,8,7,6,5,4,3	2	0.00	0.30	V	8
E2	Q1	12,13	37,11		2,8,7,6,5,4,3	9				
E3	Q1	11,13	37,12		2,9,7,6,5,4,3	8				
E4	Q1	13	37,11,12		2,9,8,6,5,4,3	7				
E5	Q1	11,12	37,13		2,9,8,7,5,4,3	6				
E6	Q1	12	37,11,13		2,9,8,7,6,4,3	5				
E7	Q1	11	37,12,13		2,9,8,7,6,5,3	4				
E8	Q1	Open	37,11, 12,13		2,9,8,7,6,5,4	3				
E9	Q2	35,34,16	37	15	38,39,40,20, 19,17,18	36				
E10	Q2	34,16	37,35		36,39,40,20, 19,17,18	38				
E11	Q2	35,16	37,34		36,38,40,20, 19,17,18	39				
E12	Q2	16	37,35,34		36,38,39,20, 19,17,18	40				
E13	Q2	35,34	37,16		36,38,39,40, 19,17,18	20				
E14	Q2	34	37,35,16		36,38,39,40, 20,17,18	19				
E15	Q2	35	37,34,16		36,38,39,40, 20,19,18	17				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

Test Condition	Parameter		Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E16	Q2	Output voltage, low level	Open	37,35, 34,16	15	36,38,39,40, 20,19,17	18	0 00	0.30	V	8
E17	Q3		24,23,22	26	21	27,28,29,30, 31,32,33	25				
E18	Q3		23,22	26,24		25,28,29,30, 31,32,33	27				
E19	Q3		24,22	26,23		25,27,29,30, 31,32,33	28				
E20	Q3		22	26,24,23		25,27,28,30, 31,32,33	29				
E21	Q3		24,23	26,22		25,27,28,29, 31,32,33	30				
E22	Q3		23	26,24,22		25,27,28,29, 30,32,33	31				
E23	Q3		24	26,23,22		25,27,28,29, 30,31,33	32				
E24	Q3		Open	26,24, 23,22		25,27,28,29, 30,31,32	33				
E25	Q1		37	Open	14	11,12,13,2	Open				
E26	Q1					12,13,9	11				
E27	Q1					11,13,8	12				
E28	Q1					13,7	11,12				
E29	Q1					11,12,6	13				
E30	Q1					12,5	11,13				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E31	Q1 Output voltage, low level	37	Open	14	11,4	12,13	0 00	0.30	V	8
E32	Q1			14	3	11,12,13				
E33	Q2			15	35,34,16,36	Open				
E34	Q2				34,16,38	35				
E35	Q2				35,16,39	34				
E36	Q2				16,40	35,34				
E37	Q2				35,34,20	16				
E38	Q2				34,19	35,16				
E39	Q2				35,17	34,16				
E40	Q2				18	35,34,16				
E41	Q3	26		21	24,23,22,25	Open				
E42	Q3				23,22,27	24				
E43	Q3				24,22,28	23				
E44	Q3				22,29	24,23				
E45	Q3				24,23,30	22				
E46	Q3				23,31	24,22				
E47	Q3				24,32	23,22				
E48	Q3				33	24,23,22				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued

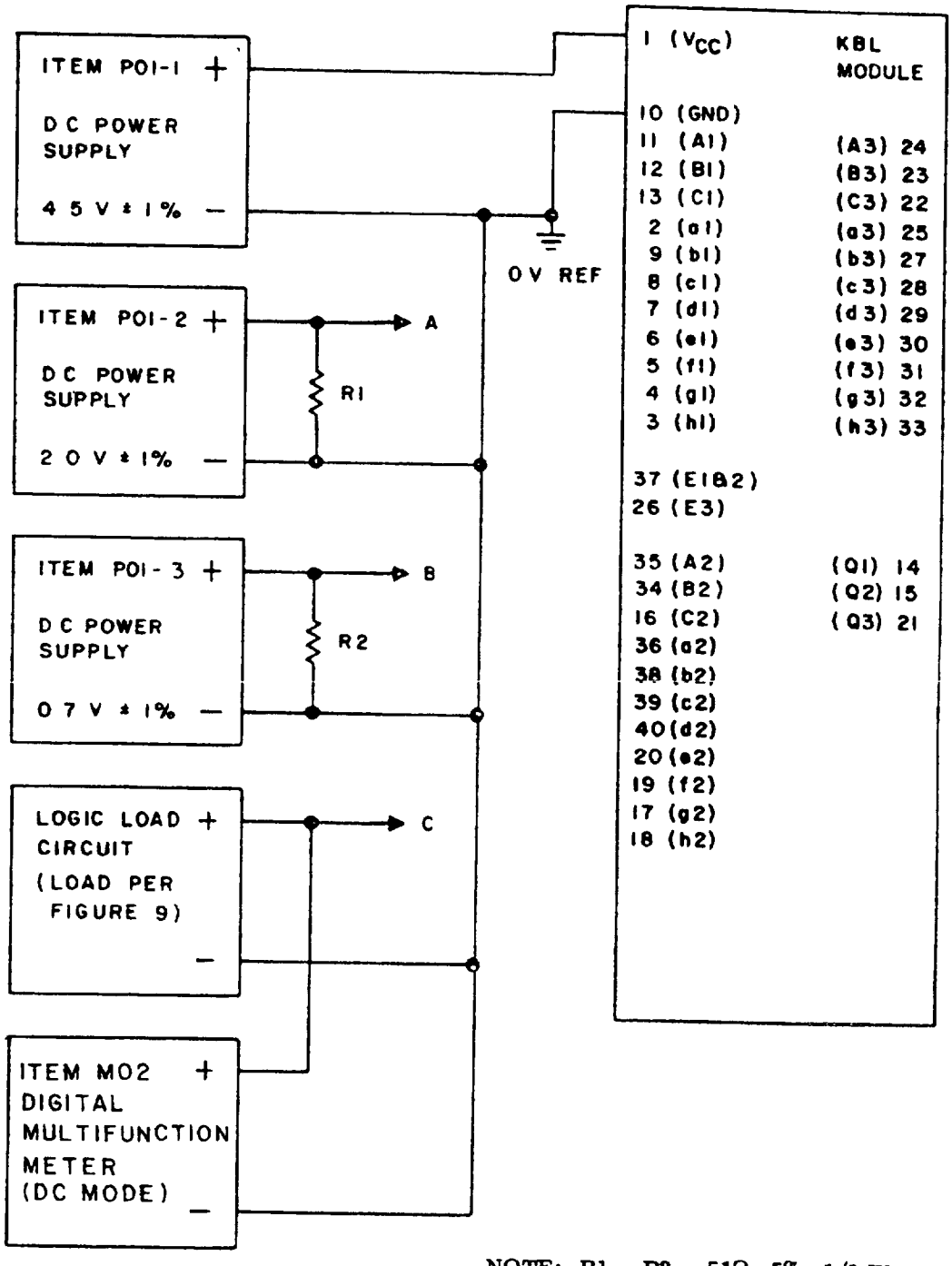
Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E49	Q1 Output voltage, low level	Open	2	14	11,12,13	37	0.00	0 30	V	8
E50	Q1		9		12,13	37,11				
E51	Q1		8		11,13	37,12				
E52	Q1		7		13	37,11,12				
E53	Q1		6		11,12	37,13				
E54	Q1		5		12	37,11,13				
E55	Q1		4		11	37,12,13				
E56	Q1		3		Open	37,11,12,13				
E57	Q2		36	15	35,34,16	37				
E58	Q2		38		34,16	37,35				
E59	Q2		39		35,16	37,34				
E60	Q2		40		16	37,35,34				
E61	Q2		20		35,34	37,16				
E62	Q2		19		34	37,35,16				
E63	Q2		17		35	37,34,16				
E64	Q2		18		Open	37,35,34,16				
E65	Q3		25	21	24,23,22	26				
E66	Q3		27	21	23,22	26,24				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (high-level and low-level) test - Continued.

Test Condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E67	Q3 Output voltage, low level	Open	28	21	24,22	26,23	0.00	0.30	V	8
E68	Q3		29		22	26,24,23				
E69	Q3		30		24,23	26,22				
E70	Q3		31		23	26,24,22				
E71	Q3		32		24	26,23,22				
E72	Q3		33		Open	26,24,23,22				

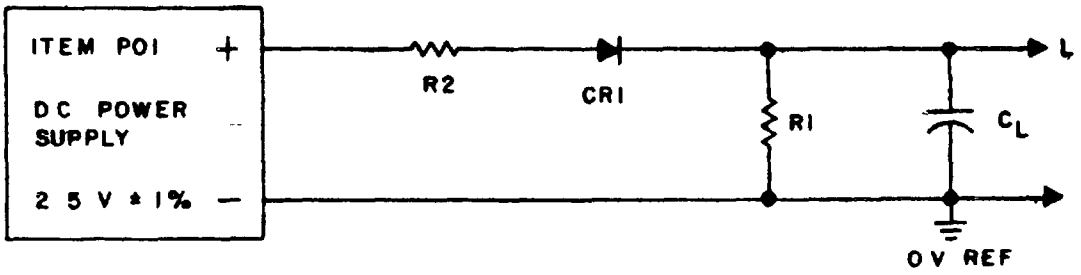
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 8. Output voltage (high-level and low-level) test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:

1. For test conditions D1-D48, E1-E72, M1-M3, N1-N3, P1-P120, Q1-Q120, S1-S3, and T1-T3:

   a. CR1 = 1N4150, or equivalent.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance).
   c. R1 = 6.04 kΩ ±1%, 1/8 W.
   d. R2 = Select value for a current flow of 4.0 mA ±5% with point L held at 0.3 volt.
   e. Capacitor C_L is not required for test conditions D and E.

2. For test conditions M4-M6, N4-N6, P121-P240, and Q121-Q240:

   a. Same as 1a
   b. C_L = 150 ±5 pF (including probe and parasitic capacitance).
   c. Same as 1c
   d. Same as 1d.

3. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
4. During output switching, the 2.5 volt power supply shall have a maximum overshoot of 500 mV peak.

FIGURE 9. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level and low-level) and output short circuit current test.

Test Condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
F1	E1 & 2 Input current, high level	37	Open	Open	11,12,13, 35,34,16	-10	40	µA	10
F2	A1	11			37,12,13		20		
F3	A2	12			37,11,13				
F4	A3	13			37,11,12				
F5	a1	2		37	11,12,13				
F6	b1	9		37,11	12,13				
F7	c1	8		37,12	11,13				
F8	d1	7		37,11,12	13				
F9	e1	6		37,13	11,12				
F10	f1	5		37,11,13	12				
F11	g1	4		37,12,13	11				
F12	h1	3		37,11,12,13	Open				
F13	A2	35		Open	37,34,16				
F14	B2	34		Open	37,35,16				
F15	C2	16		Open	37,35,34				
F16	a2	36		37	35,34,16				
F17	b2	38		37,35	34,16				
F18	c2	39		37,34	35,16				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

Test Condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
F19	d2 Input current, high level	40	Open	37,35,34	16	-10	20	µA	10
F20	e2	20		37,16	35,34				
F21	f2	19		37,35,16	34				
F22	g2	17		37,34,16	35				
F23	h2	18		37,35,34,16	Open				
F24	E3	26		Open	24,23,22				
F25	A3	24			26,23,22				
F26	B3	23			26,24,22				
F27	C3	22			26,24,23				
F28	a3	25		26	24,23,22				
F29	b3	27		26,24	23,22				
F30	c3	28		26,23	24,22				
F31	d3	29		26,24,23	22				
F32	e3	30		26,22	24,23				
F33	f3	31		26,24,22	23				
F34	g3	32		26,23,22	24				
F35	h3	33		26,24,23,22	Open				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VII Input current (high-level and low-level) and output short circuit current test - Continued

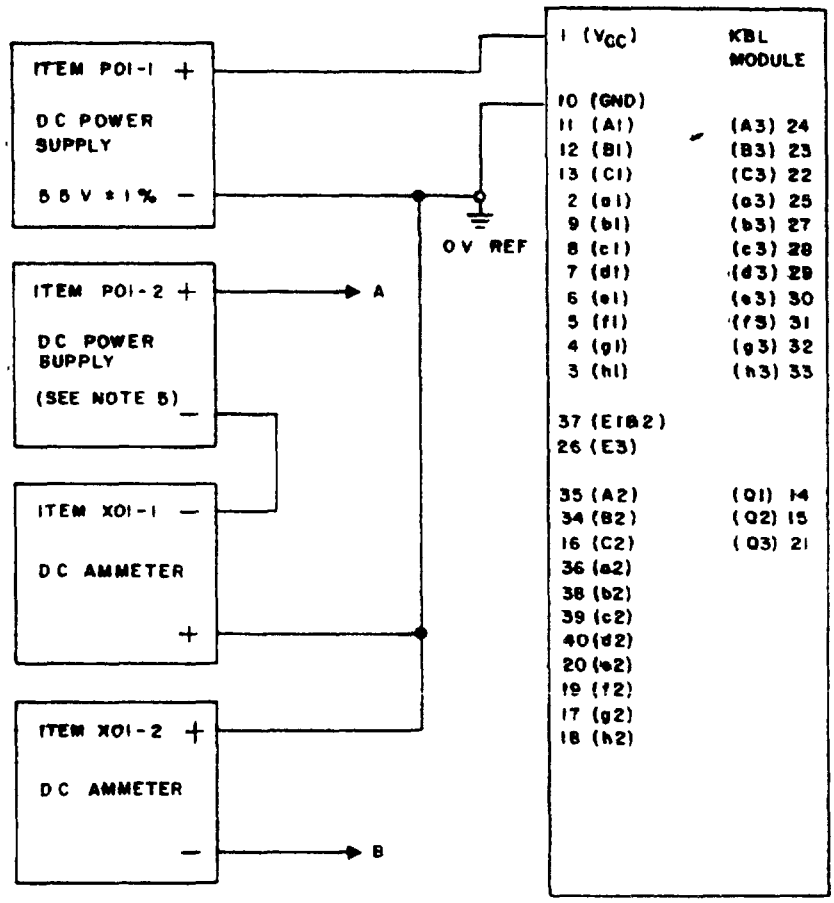
Test Condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
G1	E1 & 2 Input current, low level	11,12,13, 35,34,16	37	Open	Open	-0.01	-0.84	mA	10
G2	A1	37,12,13	11				-0.42		
G3	B1	37,11,13	12						
G4	C1	37,11,12	13						
G5	a1	11,12,13	2		37				
G6	b1	12,13	9		37,11				
G7	c1	11,13	8		37,12				
G8	d1	13	7		37,11,12				
G9	e1	11,12	6		37,13				
G10	f1	12	5		37,11,13				
G11	g1	11	4		37,12,13				
G12	h1	Open	3		37,11,12,13				
G13	A2	37,34,16	35		Open				
G14	B2	37,35,16	34		Open				
G15	C2	37,35,34	16		Open				
G16	a2	35,34,16	36		37				
G17	b2	34,16	38		37,35				
G18	c2	35,16	39		37,34				
G19	d2	16	40		37,35,34				
G20	e2	35,34	20		37,16				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued

Test Condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
G21	f2	Input current, low level	34	19	Open	37,35,16	-0.01	-0.42	mA	10
G22	g2		35	17		37,34,16				
G23	h2		Open	18		37,35,34,16				
G24	E3		24,23,22	26		Open				
G25	A3		26,23,22	24						
G26	B3		26,24,22	23						
G27	C3		26,24,23	22						
G28	a3		24,23,22	25		26				
G29	b3		23,22	27		26,24				
G30	c3		24,22	28		26,23				
G31	d3		22	29		26,24,23				
G32	e3		24,23	30		26,22				
G33	f3		23	31		26,24,22				
G34	g3		24	32		26,23,22				
G35	h3		Open	33		26,24,23,22				
H1	Q1	Output short circuit current	3	14		37,11,12,13	-5	-40		
H2	Q2	Output short circuit current	18	15		37,35,34,16	-5	-40		
H3	Q3	Output short circuit current	33	21		26,24,23,22	-5	-40		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



CAUTION.

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Read item X01-1 for test conditions F1-F35 and item X01-2 for test conditions G1-G35 and H1-H3.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. With point A disconnected, item X01-1 should indicate 0.0 μA on the 1 μA range.
4. When performing output short circuit current test, only one output shall be shorted at a time and the measurement taken within 5 seconds.
5. Test procedures.
   a. For test condition F set item P01-2 to 2.4 V ±1%.
   b. For test conditions G and H set item P01-2 to 4.5 V ±1%.

FIGURE 10. Input current (high-level and low-level) and output short circuit current test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		Pin 10 to	Min	Max		
J1	Power supply current, low level	B	10	40	mA	11
J2	Power supply current, high level	A	10	40	mA	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

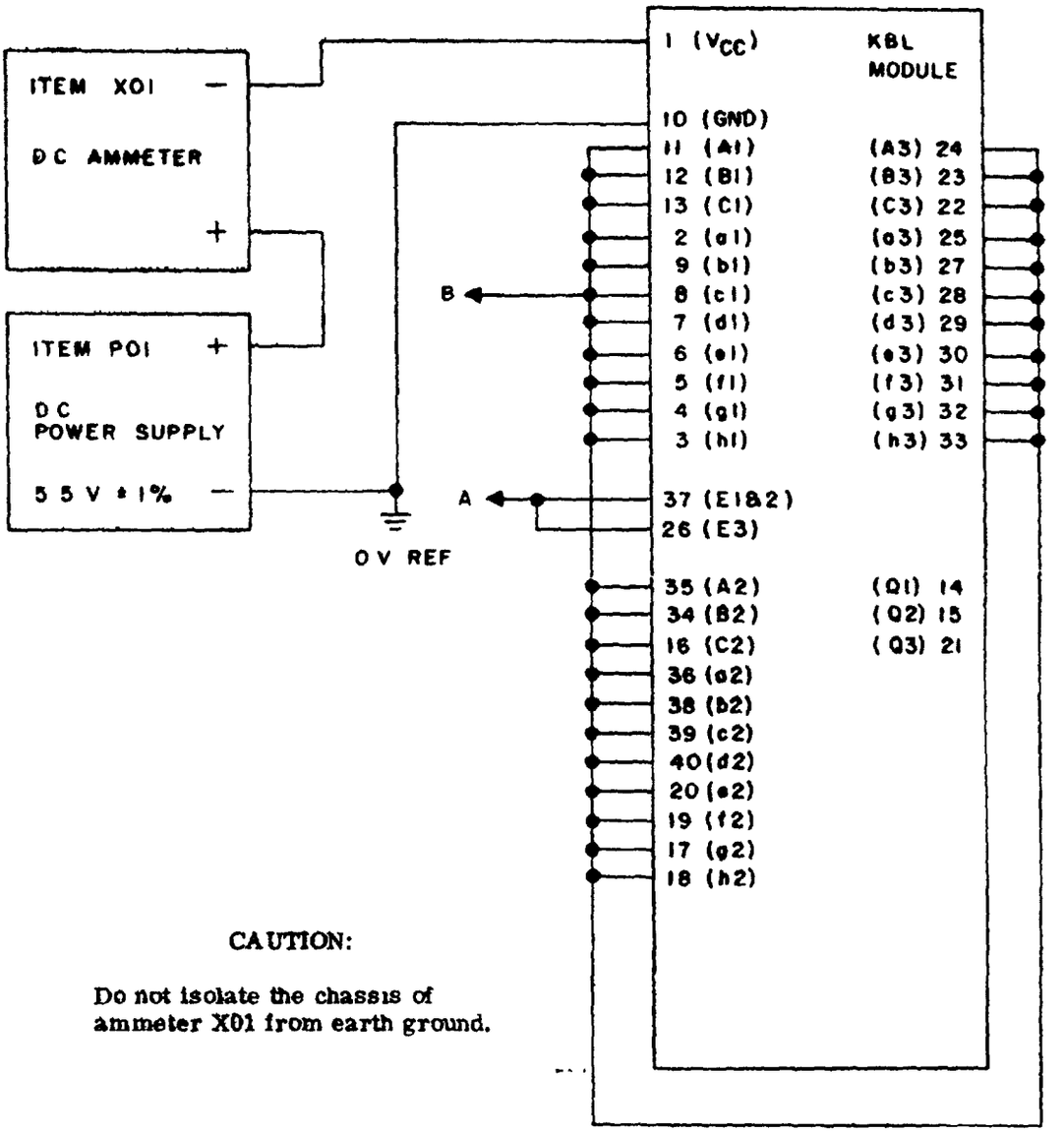


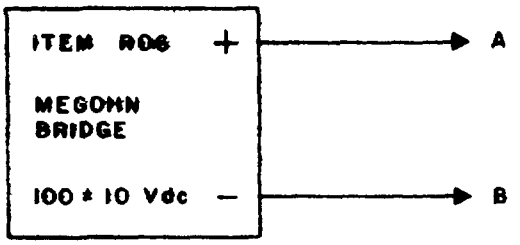
FIGURE 11. Power supply current (low-level and high-level) test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE IX. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K	Isolation	All pins	Frame	10	∞	MΩ	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



1 (V_{CC})	KBL MODULE
10 (GND)	
11 (A1)	(A3) 24
12 (B1)	(B3) 23
13 (C1)	(C3) 22
2 (a1)	(a3) 25
9 (b1)	(b3) 27
8 (c1)	(c3) 28
7 (d1)	(d3) 29
6 (e1)	(e3) 30
5 (f1)	(f3) 31
4 (g1)	(g3) 32
3 (h1)	(h3) 33
37 (E1&2)	
26 (E3)	
35 (A2)	(Q1) 14
34 (B2)	(Q2) 15
16 (C2)	(Q3) 21
36 (a2)	
38 (b2)	
39 (c2)	
40 (d2)	
20 (e2)	
19 (f2)	
17 (g2)	
18 (h2)	

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 12. Isolation test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Input capacitance test.

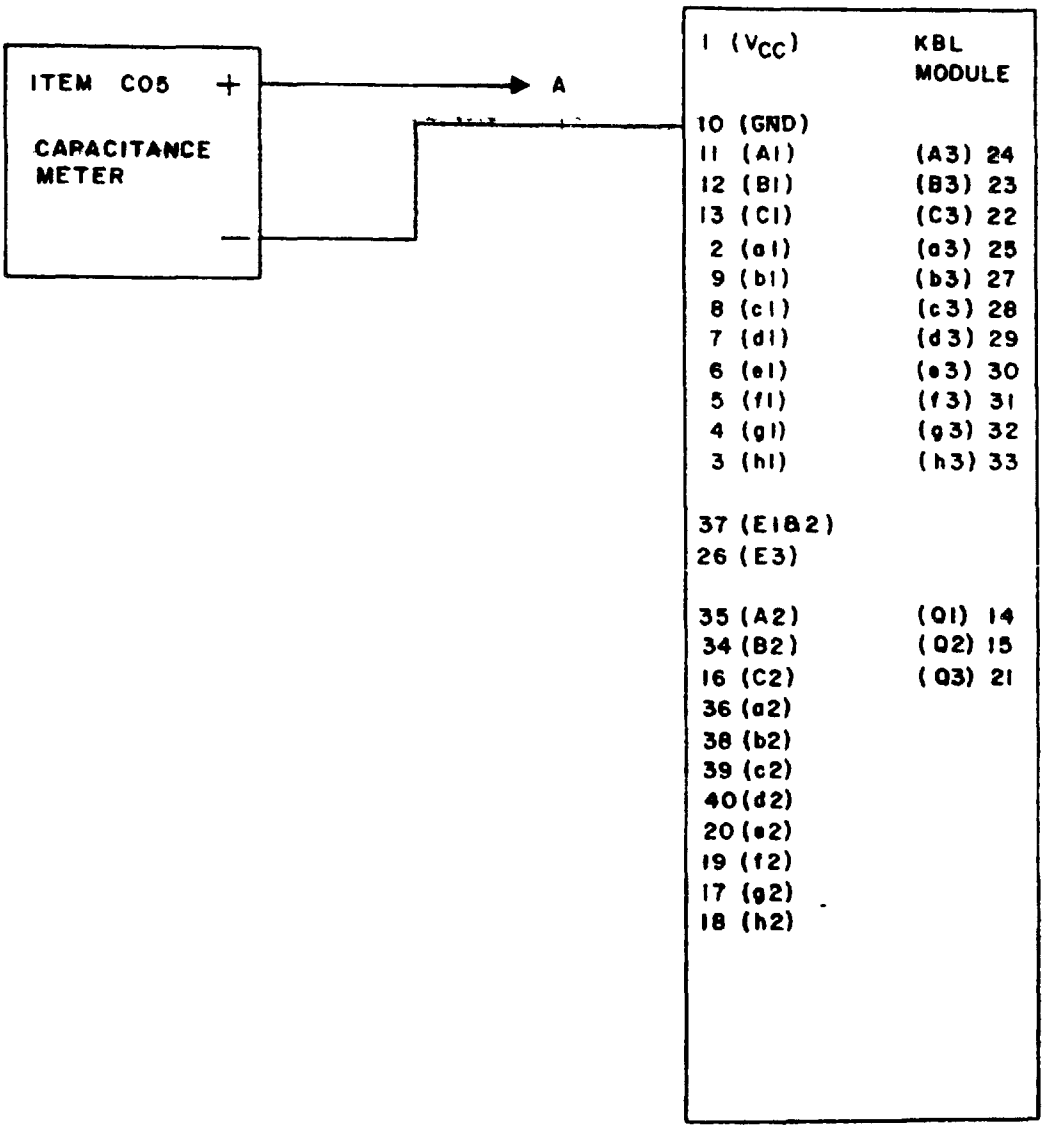
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	E1 & 2 Input capacitance	37	4	30	pF	13
L2	a1	2	2	15		
L3	b1	9				
L4	c1	8				
L5	d1	7				
L6	e1	6				
L7	f1	5				
L8	g1	4				
L9	h1	3				
L10	A1	11				
L11	B1	12				
L12	C1	13				
L13	a2	36				
L14	b2	38				
L15	c2	39				
L16	d2	40				
L17	e2	20				
L18	f2	19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L19	g2 Input capacitance	17	2	15	pF	13
L20	h2	18				
L21	A2	35				
L22	B2	34				
L23	C2	16				
L24	E3	26				
L25	a3	25				
L26	b3	27				
L27	c3	28				
L28	d3	29				
L29	e3	30				
L30	f3	31				
L31	g3	32				
L32	h3	33				
L33	A3	24				
L34	B3	23				
L35	C3	22				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 13. Input capacitance test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI Transition time and propagation delay time test.

Test condition	Parameter	Connect				Test condition M (t_{TLH}) initial limits 25° ±5°C ambient		Test condition N (t_{THL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
M1,N1	Q1 Transition time, C_L = 50 pF	2	14	11,12,13	37	5	60	2	30	ns	14,15
M2,N2	Q2 Transition time, C_L = 50 pF	36	15	35,34,16	37	5	60	2	30		
M3,N3	Q3 Transition time, C_L = 50 pF	25	21	24,23,22	26	5	60	2	30		
M4,N4	Q1 Transition time, C_L = 150 pF	2	14	11,12,13	37	10	110	5	60		
M5,N5	Q2 Transition time, C_L = 150 pF	36	15	35,34,16	37	10	110	5	60		
M6,N6	Q3 Transition time, C_L = 150 pF	25	21	24,23,22	26	10	110	5	60		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P1,Q1	a1 to Q1 Propagation delay time, C_L = 50 pF	2	14	11,12,13	37	10	95	10	85	ns	14,15
P2,Q2	b1 to Q1	9		12,13	37,11						
P3,Q3	c1 to Q1	8		11,13	37,12						
P4,Q4	d1 to Q1	7		13	37,11,12						
P5,Q5	e1 to Q1	6		11,12	37,13						
P6,Q6	f1 to Q1	5		12	37,11,13						
P7,Q7	g1 to Q1	4		11	37,12,13						
P8,Q8	h1 to Q1	3		Open	37,11,12,13						
P9,Q9	a2 to Q2	36	15	35,34,16	37						
P10,Q10	b2 to Q2	38		34,16	37,35						
P11,Q11	c2 to Q2	39		35,16	37,34						
P12,Q12	d2 to Q2	40		16	37,35,34						
P13,Q13	e2 to Q2	20		35,34	37,16						
P14,Q14	f2 to Q2	19		34	37,35,16						
P15,Q15	g2 to Q2	17		35	37,34,16						
P16,Q16	h2 to Q2	18		Open	37,35,34,16						
P17,Q17	a3 to Q3	25	21	24,23,22	26						
P18,Q18	b3 to Q3	27	21	23,22	26,24						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P19,Q19	c3 to Q3 Propagation delay time, C_L = 50 pF	28	21	24,22	26,23	10	95	10	85	ns	14,15
P20,Q20	d3 to Q3	29		22	26,24,23						
P21,Q21	e3 to Q3	30		24,23	26,22						
P22,Q22	f3 to Q3	31		23	26,24,22						
P23,Q23	g3 to Q3	32		24	26,23,22						
P24,Q24	h3 to Q3	33		Open	26,24,23,22						
P25,Q25	E1 & 2 to Q1	37	14	11,12,13,2	Open		105		105		14,16
P26,Q26	E1 & 2 to Q1			12,13,9	11						
P27,Q27	E1 & 2 to Q1			11,13,8	12						
P28,Q28	E1 & 2 to Q1			13,7	11,12						
P29,Q29	E1 & 2 to Q1			11,12,6	13						
P30,Q30	E1 & 2 to Q1			12,5	11,13						
P31,Q31	E1 & 2 to Q1			11,4	12,13						
P32,Q32	E1 & 2 to Q1			3	11,12,13						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P33,Q33	E1 & 2 to Q2 Propagation delay time, C_L = 50 pF	37	15	35,34,16,36	Open	10	105	10	105	ns	14,16
P34,Q34	E1 & 2 to Q2			34,16,38	35						
P35,Q35	E1 & 2 to Q2			35,16,39	34						
P36,Q36	E1 & 2 to Q2			16,40	35,34						
P37,Q37	E1 & 2 to Q2			35,34,20	16						
P38,Q38	E1 & 2 to Q2			34,19	35,16						
P39,Q39	E1 & 2 to Q2			35,17	34,16						
P40,Q40	E1 & 2 to Q2			18	35,34,16						
P41,Q41	E3 to Q3	26	21	24,23,22,25	Open						
P42,Q42	E3 to Q3			23,22,27	24						
P43,Q43	E3 to Q3			24,22,28	23						
P44,Q44	E3 to Q3			22,29	24,23						
P45,Q45	E3 to Q3			24,23,30	22						
P46,Q46	E3 to Q3			23,31	24,22						
P47,Q47	E3 to Q3			24,32	23,22						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter		Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P48,Q48	E3 to Q3	Propagation delay time, C_L = 50 pF	26	21	33	24,23,22	10	105	10	105	ns	14,16
P49,Q49	A1 to Q1		11	14	12,13,2	37,9		110		115		14,15
P50,Q50	A1 to Q1				12,13,9	37,2						14,16
P51,Q51	A1 to Q1				13,8	37,12,7						14,15
P52,Q52	A1 to Q1				13,7	37,12,8						14,16
P53,Q53	A1 to Q1				12,6	37,13,5						14,15
P54,Q54	A1 to Q1				12,5	37,13,6						14,16
P55,Q55	A1 to Q1				4	37,12,13,3						14,15
P56,Q56	A1 to Q1				3	37,12,13,4						14,16
P57,Q57	B1 to Q1		12		11,13,2	37,8						14,15
P58,Q58	B1 to Q1				11,13,8	37,2						14,16
P59,Q59	B1 to Q1				13,9	37,11,7						14,15
P60,Q60	B1 to Q1				13,7	37,11,9						14,16
P61,Q61	B1 to Q1				11,6	37,13,4						14,15
P62,Q62	B1 to Q1				11,4	37,13,6						14,16
P63,Q63	B1 to Q1				5	37,11,13,3						14,15
P64,Q64	B1 to Q1				3	37,11,13,5						14,16
P65,Q65	C1 to Q1		13		11,12,2	37,6						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter		Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P66,Q66	C1 to Q1	Propagation delay time, C_L = 50 pF	13	14	11,12,6	37,2	10	110	10	115	ns	14,16
P67,Q67	C1 to Q1				12,9	37,11,5						14,15
P68,Q68	C1 to Q1				12,5	37,11,9						14,16
P69,Q69	C1 to Q1				11,8	37,12,4						14,15
P70,Q70	C1 to Q1				11,4	37,12,8						14,16
P71,Q71	C1 to Q1				7	37,11,12,3						14,15
P72,Q72	C1 to Q1				3	37,11,12,7						14,16
P73,Q73	A2 to Q2		35	15	34,16,36	37,38						14,15
P74,Q74	A2 to Q2				34,16,38	37,36						14,16
P75,Q75	A2 to Q2				16,39	37,34,40						14,15
P76,Q76	A2 to Q2				16,40	37,34,39						14,16
P77,Q77	A2 to Q2				34,20	37,16,19						14,15
P78,Q78	A2 to Q2				34,19	37,16,20						14,16
P79,Q79	A2 to Q2				17	37,34,16,18						14,15
P80,Q80	A2 to Q2				18	37,34,16,17						14,16
P81,Q81	B2 to Q2		34		35,16,36	37,39						14,15
P82,Q82	B2 to Q2		34		35,16,39	37,36						14,16
P83,Q83	B2 to Q2		34		16,38	37,35,40						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P84,Q84	B2 to Q2 Propagation delay time, C_L = 50 pF	34	15	16,40	37,35,38	10	110	10	115	ns	14,16
P85,Q85	B2 to Q2			35,20	37,16,17						14,15
P86,Q86	B2 to Q2			35,17	37,16,20						14,16
P87,Q87	B2 to Q2			19	37,35,16,18						14,15
P88,Q88	B2 to Q2			18	37,35,16,19						14,16
P89,Q89	C2 to Q2	16		35,34,36	37,20						14,15
P90,Q90	C2 to Q2			35,34,20	37,36						14,16
P91,Q91	C2 to Q2			34,38	37,35,19						14,15
P92,Q92	C2 to Q2			34,19	37,35,38						14,16
P93,Q93	C2 to Q2			35,39	37,34,17						14,15
P94,Q94	C2 to Q2			35,17	37,34,39						14,16
P95,Q95	C2 to Q2			40	37,35,34,18						14,15
P96,Q96	C2 to Q2			18	37,35,34,40						14,16
P97,Q97	A3 to Q3	24	21	23,22,25	26,27						14,15
P98,Q98	A3 to Q3			23,22,27	26,25						14,16
P99,Q99	A3 to Q3			22,28	26,23,29						14,15
P100,Q100	A3 to Q3			22,29	26,23,28						14,16
P101,Q101	A3 to Q3			23,30	26,22,31						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P102,Q102	A3 to Q3 Propagation delay time, C_L = 50 pF	24	21	23,31	26,22,30	10	110	10	115	ns	14,16
P103,Q103	A3 to Q3	24		32	26,23,22,33						14,15
P104,Q104	A3 to Q3	24		33	26,23,22,32						14,16
P105,Q105	B3 to Q3	23		24,22,25	26,28						14,15
P106,Q106	B3 to Q3			24,22,28	26,25						14,16
P107,Q107	B3 to Q3			22,27	26,24,29						14,15
P108,Q108	B3 to Q3			22,29	26,24,27						14,16
P109,Q109	B3 to Q3			24,30	26,22,32						14,15
P110,Q110	B3 to Q3			24,32	26,22,30						14,16
P111,Q111	B3 to Q3			31	26,24,22,33						14,15
P112,Q112	B3 to Q3			33	26,24,22,31						14,16
P113,Q113	C3 to Q3	22		24,23,25	26,30						14,15
P114,Q114	C3 to Q3			24,23,30	26,25						14,16
P115,Q115	C3 to Q3			23,27	26,24,31						14,15
P116,Q116	C3 to Q3			23,31	26,24,27						14,16
P117,Q117	C3 to Q3			24,28	26,23,32						14,15
P118,Q118	C3 to Q3			24,32	26,23,28						14,16
P119,Q119	C3 to Q3			29	26,24,23,33						14,15
P120,Q120	C3 to Q3			33	26,24,23,29						14,16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P121,Q121	a1 to Q1	2	14	11,12,13	37	10	110	10	100	ns	14,15
P122,Q122	b1 to Q1	9		12,13	37,11						
P123,Q123	c1 to Q1	8		11,13	37,12						
P124,Q124	d1 to Q1	7		13	37,11,12						
P125,Q125	e1 to Q1	6		11,12	37,13						
P126,Q126	f1 to Q1	5		12	37,11,13						
P127,Q127	g1 to Q1	4		11	37,12,13						
P128,Q128	h1 to Q1	3		Open	37,11,12,13						
P129,Q129	a2 to Q2	36	15	35,34,16	37						
P130,Q130	b2 to Q2	38		34,16	37,35						
P131,Q131	c2 to Q2	39		35,16	37,34						
P132,Q132	d2 to Q2	40		16	37,35,34						
P133,Q133	e2 to Q2	20		35,34	37,16						
P134,Q134	f2 to Q2	19		34	37,35,16						
P135,Q135	g2 to Q2	17		35	37,34,16						
P136,Q136	h2 to Q2	18		Open	37,35,34,16						
P137,Q137	a3 to Q3	25	21	24,23,22	26						
P138,Q138	b3 to Q3	27	21	23,22	26,24						

Propagation delay time, C_L = 150 pF

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P139,Q139	c3 to Q3 Propagation delay time, C_L = 150 pF	28	21	24,22	26,23	10	110	10	100	ns	14,15
P140,Q140	d3 to Q3	29		22	26,24,23						
P141,Q141	e3 to Q3	30		24,23	26,22						
P142,Q142	f3 to Q3	31		23	26,24,22						
P143,Q143	g3 to Q3	32		24	26,23,22						
P144,Q144	h3 to Q3	33		Open	26,24,23,22						
P145,Q145	E1 & 2 to Q1	37	14	11,12,13,2	Open		120		120		14,16
P146,Q146	E1 & 2 to Q1			12,13,9	11						
P147,Q147	E1 & 2 to Q1			11,13,8	12						
P148,Q148	E1 & 2 to Q1			13,7	11,12						
P149,Q149	E1 & 2 to Q1			11,12,6	13						
P150,Q150	E1 & 2 to Q1			12,5	11,13						
P151,Q151	E1 & 2 to Q1			11,4	12,13						
P152,Q152	E1 & 2 to Q1			3	11,12,13						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P153,Q153	E1 & 2 to Q2 Propagation delay time, C_L = 150 pF	37	15	35,34,16,36	Open	10	120	10	120	ns	14,16
P154,Q154	E1 & 2 to Q2			34,16,38	35						
P155,Q155	E1 & 2 to Q2			35,16,39	34						
P156,Q156	E1 & 2 to Q2			16,40	35,34						
P157,Q157	E1 & 2 to Q2			35,34,20	16						
P158,Q158	E1 & 2 to Q2			34,19	35,16						
P159,Q159	E1 & 2 to Q2			35,17	34,16						
P160,Q160	E1 & 2 to Q2			18	35,34,16						
P161,Q161	E3 to Q3	26	21	24,23,22,25	Open						
P162,Q162	E3 to Q3			23,22,27	24						
P163,Q163	E3 to Q3			24,22,28	23						
P164,Q164	E3 to Q3			22,29	24,23						
P165,Q165	E3 to Q3			24,23,30	22						
P166,Q166	E3 to Q3			23,31	24,22						
P167,Q167	E3 to Q3			24,32	23,22						

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P168,Q168	E3 to Q3 Propagation delay time, C_L = 150 pF	26	21	33	24,23,22	10	120	10	120	ns	14,16
P169,Q169	A1 to Q1	11	14	12,13,2	37,9		125		130		14,15
P170,Q170	A1 to Q1			12,13,9	37,2						14,16
P171,Q171	A1 to Q1			13,8	37,12,7						14,15
P172,Q172	A1 to Q1			13,7	37,12,8						14,16
P173,Q173	A1 to Q1			12,6	37,13,5						14,15
P174,Q174	A1 to Q1			12,5	37,13,6						14,16
P175,Q175	A1 to Q1			4	37,12,13,3						14,15
P176,Q176	A1 to Q1			3	37,12,13,4						14,16
P177,Q177	B1 to Q1	12		11,13,2	37,8						14,15
P178,Q178	B1 to Q1			11,13,8	37,2						14,16
P179,Q179	B1 to Q1			13,9	37,11,7						14,15
P180,Q180	B1 to Q1			13,7	37,11,9						14,16
P181,Q181	B1 to Q1			11,6	37,13,4						14,15
P182,Q182	B1 to Q1			11,4	37,13,6						14,16
P183,Q183	B1 to Q1			5	37,11,13,3						14,15
P184,Q184	B1 to Q1			3	37,11,13,5						14,16
P185,Q185	C1 to Q1	13		11,12,2	37,6						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI Transition time and propagation delay time test - Continued.

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P186,Q186	C1 to Q1 Propagation delay time, C_L = 150 pF	13	14	11,12,6	37,2	10	125	10	130	ns	14,16
P187,Q187	C1 to Q1			12,9	37,11,5						14,15
P188,Q188	C1 to Q1			12,5	37,11,9						14,16
P189,Q189	C1 to Q1			11,8	37,12,4						14,15
P190,Q190	C1 to Q1			11,4	37,12,8						14,16
P191,Q191	C1 to Q1			7	37,11,12,3						14,15
P192,Q192	C1 to Q1			3	37,11,12,7						14,16
P193,Q193	A2 to Q2	35	15	34,16,36	37,38						14,15
P194,Q194	A2 to Q2			34,16,38	37,36						14,16
P195,Q195	A2 to Q2			16,39	37,34,40						14,15
P196,Q196	A2 to Q2			16,40	37,34,39						14,16
P197,Q197	A2 to Q2			34,20	37,16,19						14,15
P198,Q198	A2 to Q2			34,19	37,16,20						14,16
P199,Q199	A2 to Q2			17	37,34,16,18						14,15
P200,Q200	A2 to Q2			18	37,34,16,17						14,16
P201,Q201	B2 to Q2	34		35,16,36	37,39						14,15
P202,Q202	B2 to Q2	34		35,16,39	37,36						14,16
P203,Q203	B2 to Q2	34		16,38	37,35,40						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued.

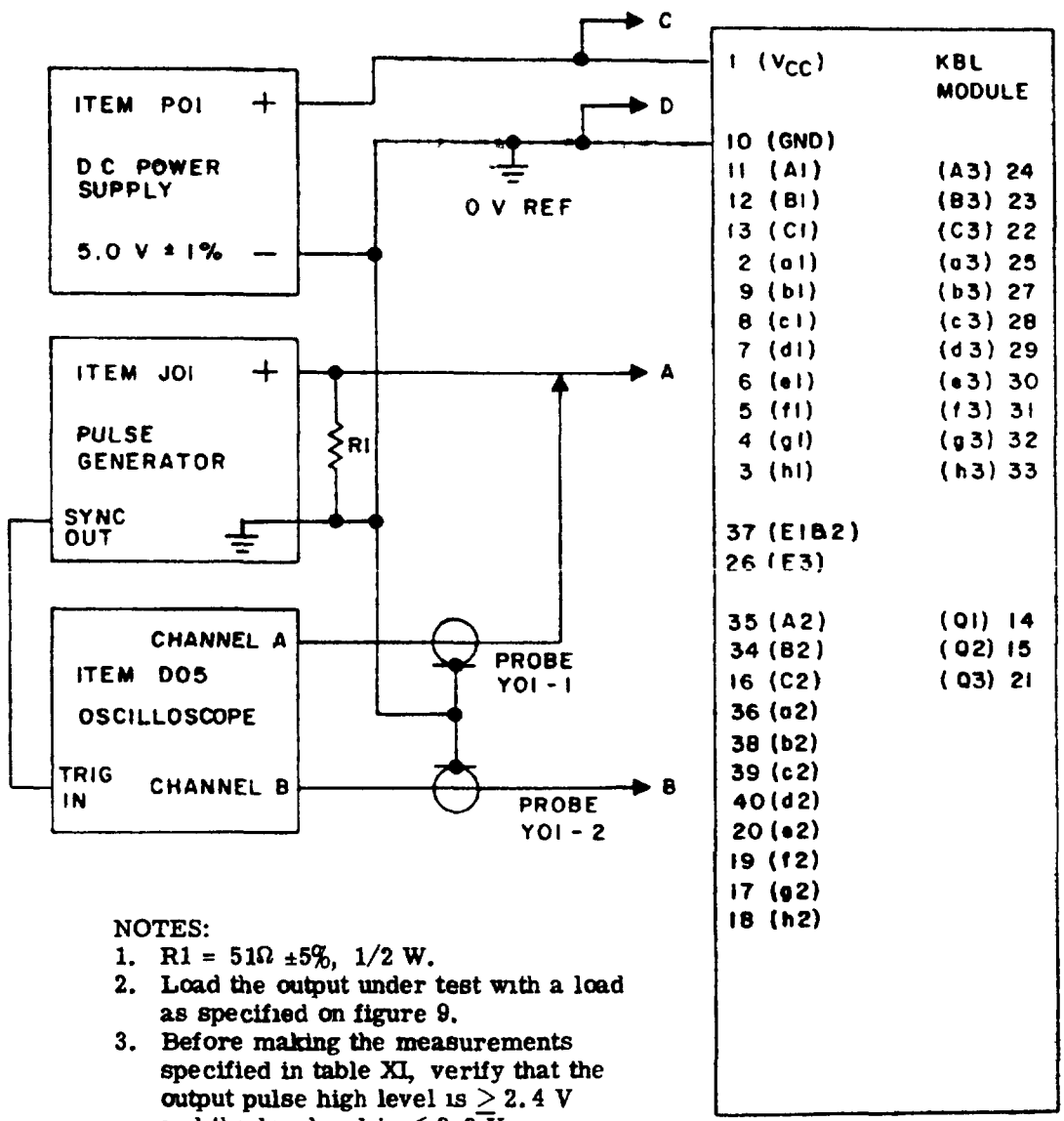
Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P204,Q204	B2 to Q2 Propagation delay time, C_L = 150 pF	34	15	16,40	37,35,38	10	125	10	130	ns	14,16
P205,Q205	B2 to Q2			35,20	37,16,17						14,15
P206,Q206	B2 to Q2			35,17	37,16,20						14,16
P207,Q207	B2 to Q2			19	37,35,16,18						14,15
P208,Q208	B2 to Q2			18	37,35,16,19						14,16
P209,Q209	C2 to Q2	16		35,34,36	37,20						14,15
P210,Q210	C2 to Q2			35,34,20	37,36						14,16
P211,Q211	C2 to Q2			34,38	37,35,19						14,15
P212,Q212	C2 to Q2			34,19	37,35,38						14,16
P213,Q213	C2 to Q2			35,39	37,34,17						14,15
P214,Q214	C2 to Q2			35,17	37,34,39						14,16
P215,Q215	C2 to Q2			40	37,35,34,18						14,15
P216,Q216	C2 to Q2			18	37,35,34,40						14,16
P217,Q217	A3 to Q3	24	21	23,22,25	26,27						14,15
P218,Q218	A3 to Q3			23,22,27	26,25						14,16
P219,Q219	A3 to Q3			22,28	26,23,29						14,15
P220,Q220	A3 to Q3			22,29	26,23,28						14,16
P221,Q221	A3 to Q3			23,30	26,22,31						14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time and propagation delay time test - Continued

Test condition	Parameter	Connect				Test condition P (t_{PLH}) initial limits 25° ±5°C ambient		Test condition Q (t_{PHL}) initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
P222,Q222	A3 to Q3 Propagation delay time, C_L = 150 pF	24	21	23,31	26,22,30	10	125	10	130	ns	14,16
P223,Q223	A3 to Q3	24		32	26,23,22,33						14,15
P224,Q224	A3 to Q3	24		33	26,23,22,32						14,16
P225,Q225	B3 to Q3	23		24,22,25	26,28						14,15
P226,Q226	B3 to Q3			24,22,28	26,25						14,16
P227,Q227	B3 to Q3			22,27	26,24,29						14,15
P228,Q228	B3 to Q3			22,29	26,24,27						14,16
P229,Q229	B3 to Q3			24,30	26,22,32						14,15
P230,Q230	B3 to Q3			24,32	26,22,30						14,16
P231,Q231	B3 to Q3			31	26,24,22,33						14,15
P232,Q232	B3 to Q3			33	26,24,22,31						14,16
P233,Q233	C3 to Q3	22		24,23,25	26,30						14,15
P234,Q234	C3 to Q3			24,23,30	26,25						14,16
P235,Q235	C3 to Q3			23,27	26,24,31						14,15
P236,Q236	C3 to Q3			23,31	26,24,27						14,16
P237,Q237	C3 to Q3			24,28	26,23,32						14,15
P238,Q238	C3 to Q3			24,32	26,23,28						14,16
P239,Q239	C3 to Q3			29	26,24,23,33						14,15
P240,Q240	C3 to Q3			33	26,24,23,29						14,16

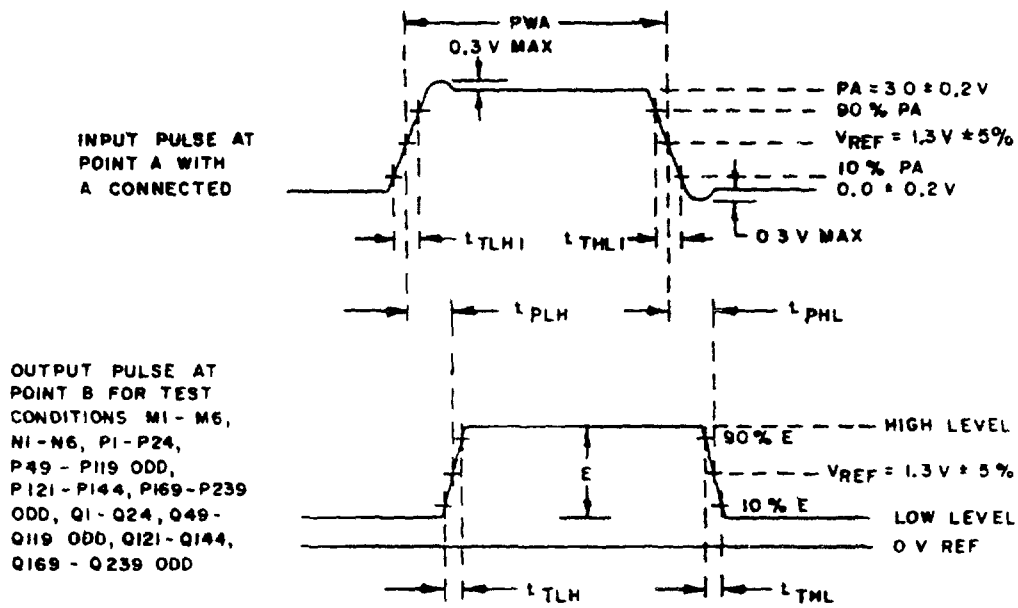
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load the output under test with a load as specified on figure 9.
3. Before making the measurements specified in table XI, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.

FIGURE 14. Transition time and propagation delay time test circuit for table XI.

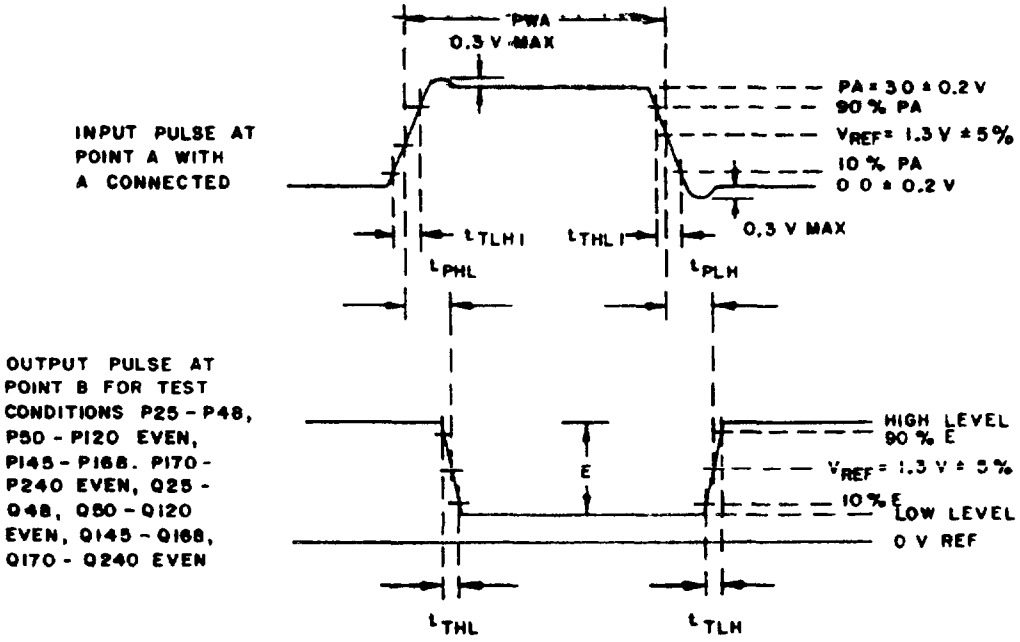
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
PRR = 1.0 ±0.1 MHz.

FIGURE 15. Transition time and propagation delay time test circuit waveforms for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
• Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
$t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
PRR = 1.0 ±0.1 MHz.

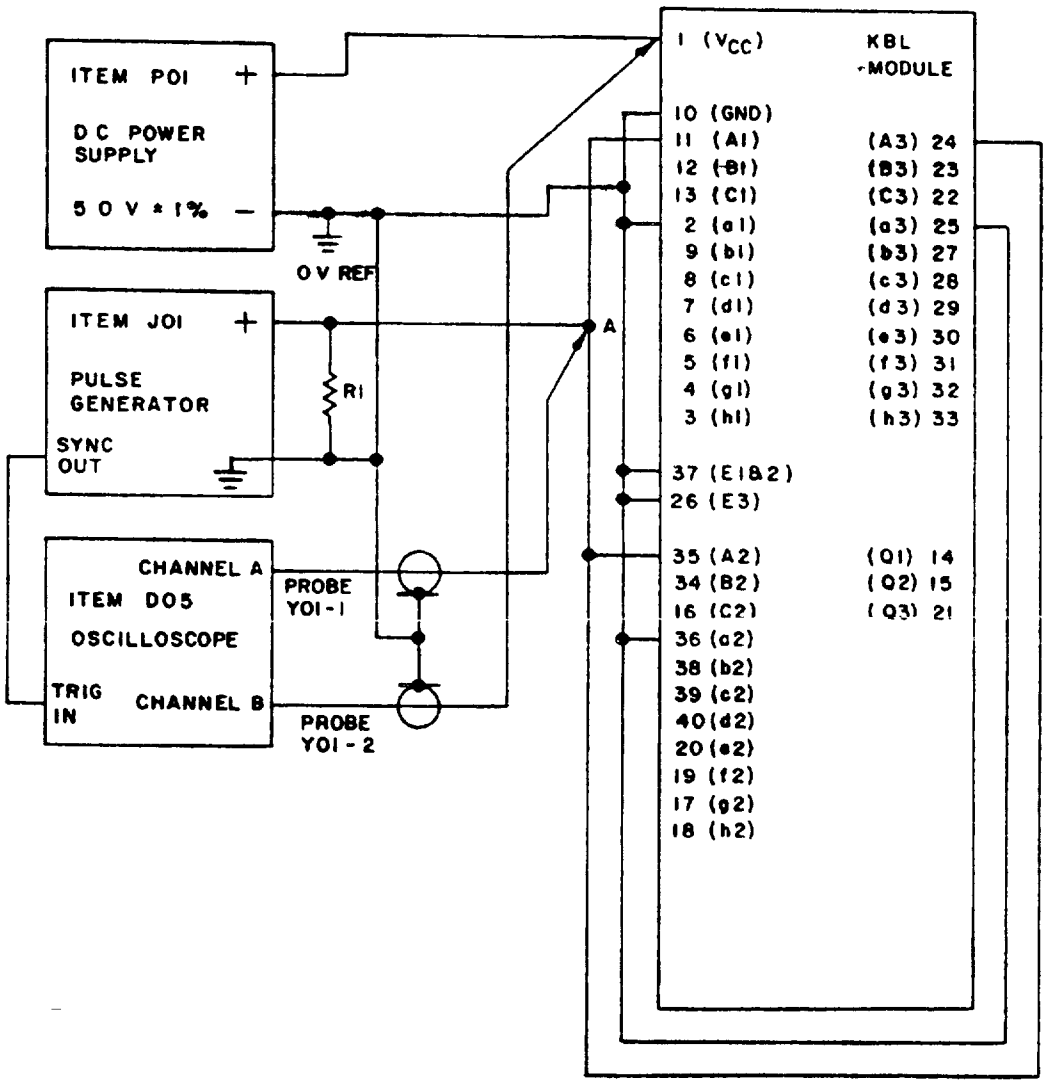
FIGURE 16. Propagation delay time test circuit waveforms for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	17,18

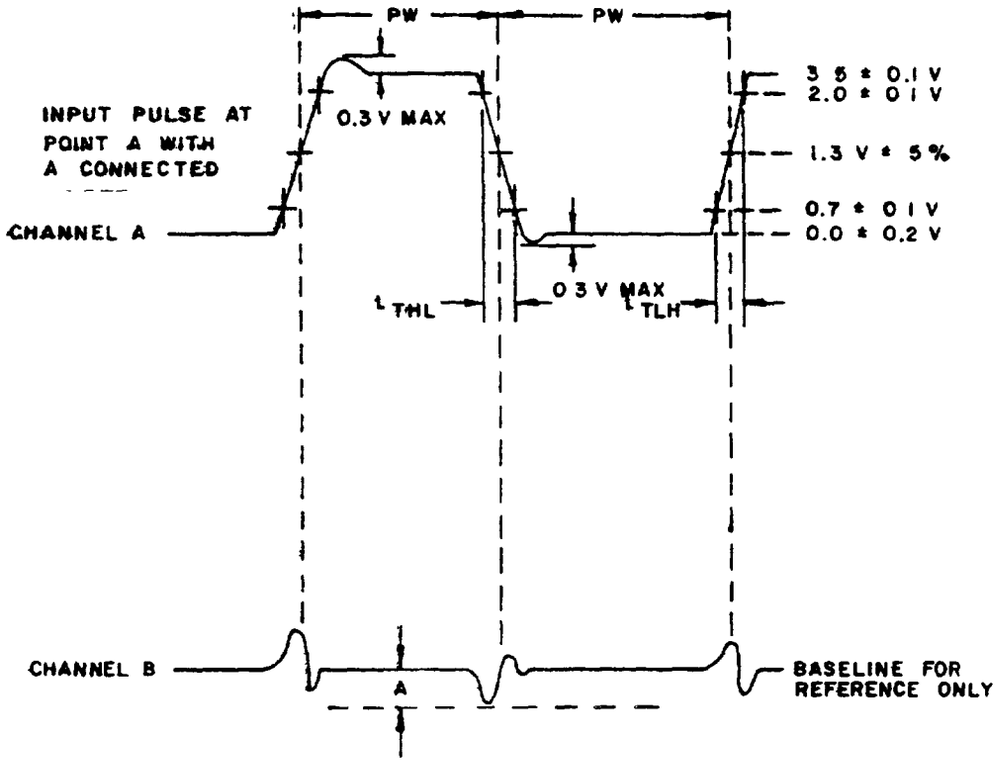
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 17. Power supply transient amplitude test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   a. PW = 500 ±50 ns.
   b. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
2. Value A is measured from the baseline to the most negative excursion of the waveform.

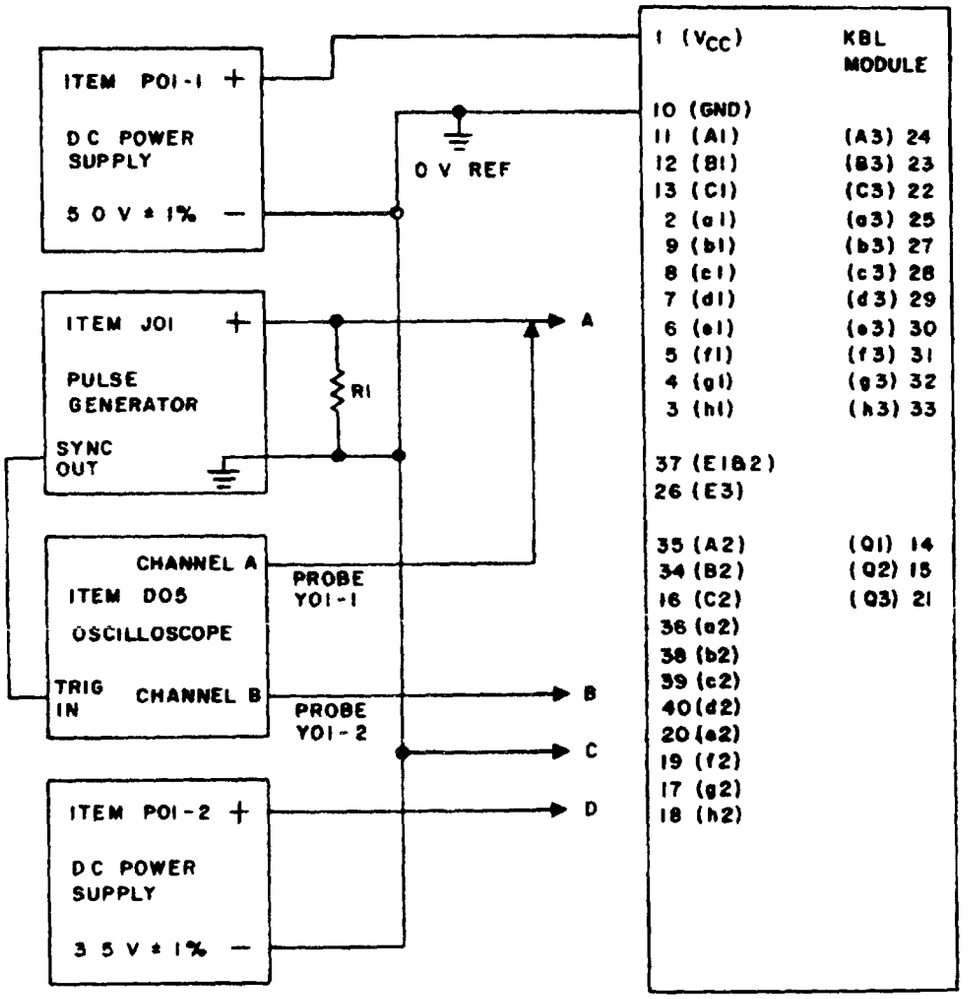
FIGURE 18. Power supply transient amplitude test circuit waveforms for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output crosstalk (high-level and low-level) test

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Q1 Output crosstalk, high level	36,25	14	37,26	11,12,13,2	0	400	mV	19,20
S2	Q2 Output crosstalk, high level	2,25	15	37,26	35,34,16,36	↓	↓	↓	↓
S3	Q3 Output crosstalk, high level	2,36	21	37,26	24,23,22,25				
T1	Q1 Output crosstalk, low level	36,25	14	37,26,2	11,12,13				
T2	Q2 Output crosstalk, low level	2,25	15	37,26,36	35,34,16				
T3	Q3 Output crosstalk, low level	2,36	21	37,26,25	24,23,22				

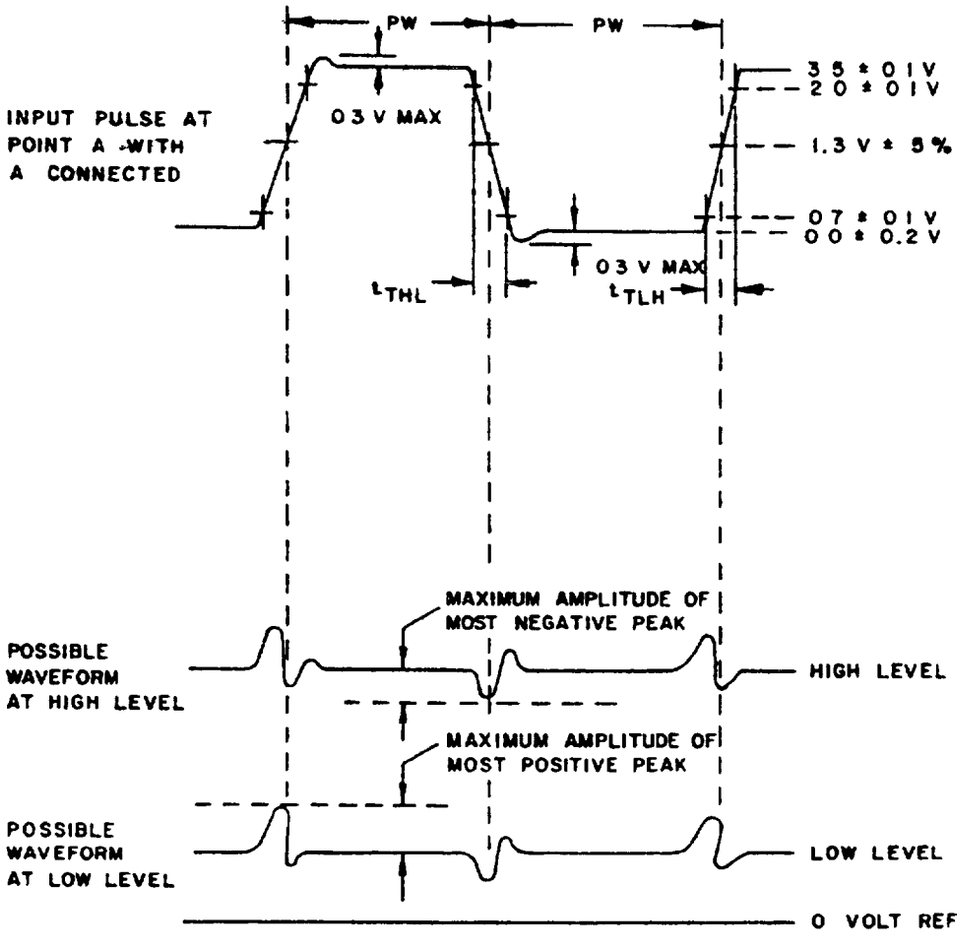
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 9.

FIGURE 19 Output crosstalk (high-level and low-level) test circuit for table XIII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).

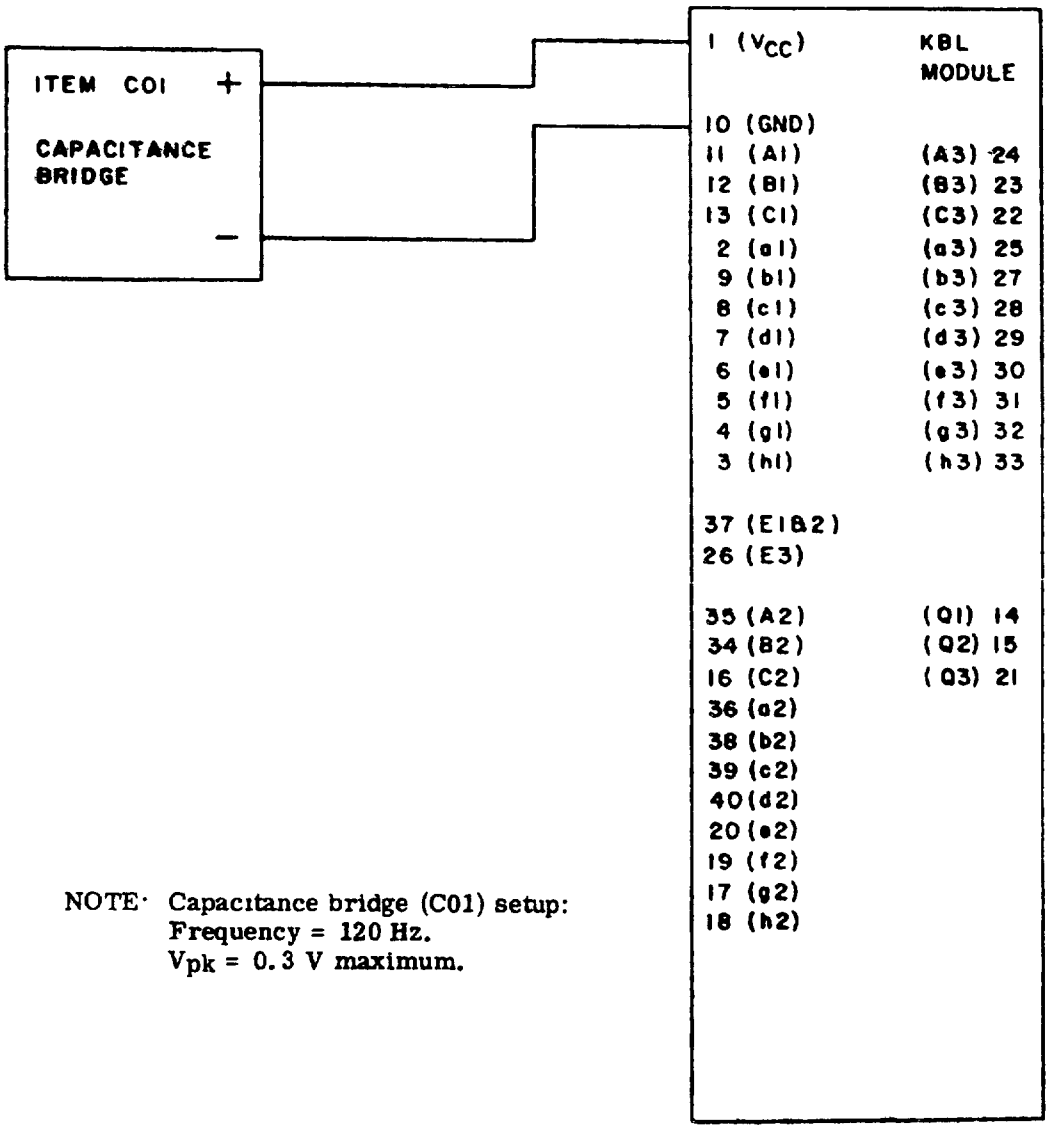
FIGURE 20. Output crosstalk (high-level and low-level) test circuit waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply filter capacitance test

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply filter capacitance	1.6	3.0	µF	21

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Capacitance bridge (C01) setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 21. Power supply filter capacitance test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Attn; ELEX 8111
Washington, DC 20360

OLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE: MIL-M-28787/40B(EC) MODULES, STANDARD ELECTRONIC MULTIPLEXER, DIGITAL, KEY CODE KBL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)	TELEPHONE NO
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.